VORON



DOWNDRAFT ASSEMBLY

We build space shuttles with gardening tools
so anyone can have a space shuttle of their own.

VERSION 2026-08-15



Before you begin on your journey, a word of caution:

You are about to assemble a robot in the comfort of your own home. This machine can maim, burn, and electrocute you if you are not careful. Please do not become the first VORON fatality. There is no special Reddit flair for that.

Please read the entire manual before you start assembly. As you begin wrenching, please check our Discord channels for any tips and questions that may halt your progress.

Most of all, good luck!

THE VORON TEAM

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PART PRINTING GUIDELINES

The VORON Team has provided the following print guidelines to follow for the best chance at success with your parts. There are often questions about substituting materials or changing printing standards, but we recommend that you follow these:

3D PRINTING PROCESS

Fused Deposition Modeling (FDM)

MATERIAL

ABS or ASA

LAYER HEIGHT

Recommended: 0.2mm

EXTRUSION WIDTH

Recommended: Forced 0.4mm

SLICING METHOD

Recommended: Classic

INFILL TYPE

Grid, Gyroid, Honeycomb, Triangle or Cubic

INFILL PERCENTAGE

Recommended: 40%

WALL COUNT

Recommended: 4

SOLID TOP/BOTTOM LAYERS

Recommended: 5

FILE NAMING

By this time you should have already downloaded our STL files from the VORON GitHub. You might have noticed that we have used a unique naming convention for the files. This is how to use them.

PRIMARY COLOR

Example z_joint_lower_x4.stl

These files will have nothing at the start of the filename.

ACCENT COLOR

Example [a]_tensioner_left.stl

We have added “[a]” to the front of any STL file that is intended to be printed with accent color.

CLEAR/TRANSPARENT

Example [c]_led_diffuser.stl

We have added “[c]” to the front of any STL file that is intended to be printed with clear or transparent filament (cloudy is better than clear; white or light grey may also work in a pinch).

OPAQUE/SOLID

Example [o]_led_diffuser_mask.stl

We have added “[o]” to the front of any STL file that is intended to be printed with an opaque filament (blocks light, dark colours and blacks).

QUANTITY REQUIRED

Example din_frame_mount_x4.stl

If any file ends with “_x#”, that is telling you the quantity of that part required to build the machine.

HOW TO GET HELP

If you need assistance with your build, we’re here to help. Head on over to our Discord group and post your questions. This is our primary medium to help VORON Users and we have a great community that can help you out if you get stuck.

DISCORD https://discord.gg/voron

VORON FORUM https://forum.vorondesign.com

REPORTING ISSUES

Should you find an issue in the documentation or have a suggestion for an improvement please consider opening an issue on GitHub (https://github.com/VoronDesign/Downdraft/issues).
When raising an issue please include the relevant page numbers and a short description; annotated screenshots are also very welcome. We periodically update the manual based on the feedback we get.

THIS IS JUST A REFERENCE

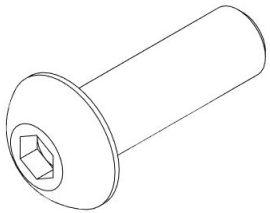
This manual is designed to be a simple reference manual. Building a VORON can be a complex endeavour and for that reason we recommend downloading the CAD files off our Github repository if there are sections you need clarification on. It can sometimes be easier to follow along when you have the whole assembly in front of you.



https://github.com/vorondesign



https://docs.vorondesign.com



BUTTON HEAD CAP SCREW (BHCS)

Metric fastener with a domed shape head and hex drive. Most commonly found in locations where M5 fasteners are used.

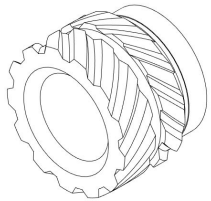
ISO 7380-1



FLAT HEAD COUNTERSUNK SCREW (FHCS)

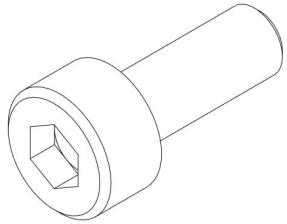
Metric fastener with a cone shaped head and a flat top.

ISO 10642



HEATSET INSERT

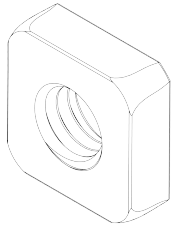
Use a soldering iron to heat the M3 insert and melt it into the plastic as it's pressed in. As the plastic cools, it holds on to the knurls and ridges, giving excellent resistance to both torque and pull-out.



SOCKET HEAD CAP SCREW (SHCS)

Metric fastener with a cylindrical head and hex drive. The most common fastener used on the VORON.

ISO 4762



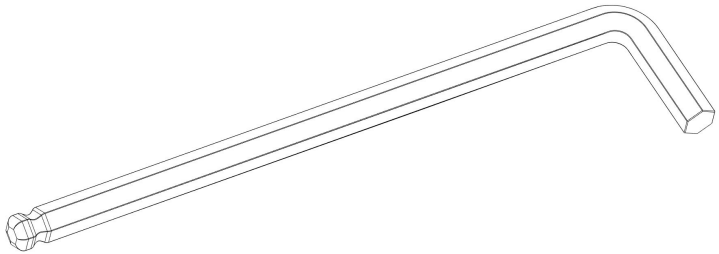
M3 SQUARE NUT

Nut that is used where parts need to be tightly held together, often where headset inserts would not be suitable.

DIN 562

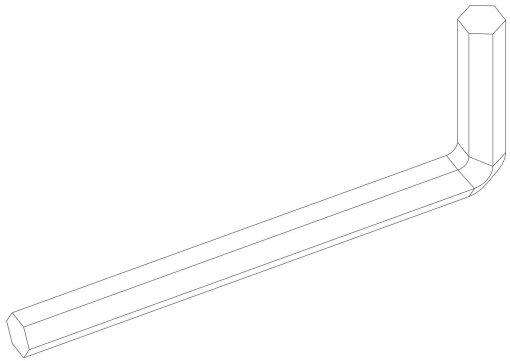
BALL-END DRIVER

Some parts of this design require the use of a ball-end hex driver for assembly. We recommend you get 2.0 mm, 2.5 mm, and 3.0 mm ones.



2.5MM HEX DRIVER

The 2.5mm hex driver will see a lot of use in this build. A quality driver is strongly recommended. Refer to the sourcing guide for suggestions.



ADDITIONAL TOOLS

We provide additional tool recommendations in our sourcing guide. Visit https://vorondesign.com/sourcing_guide and switch to the “VORON Tools” tab at the bottom of the page.

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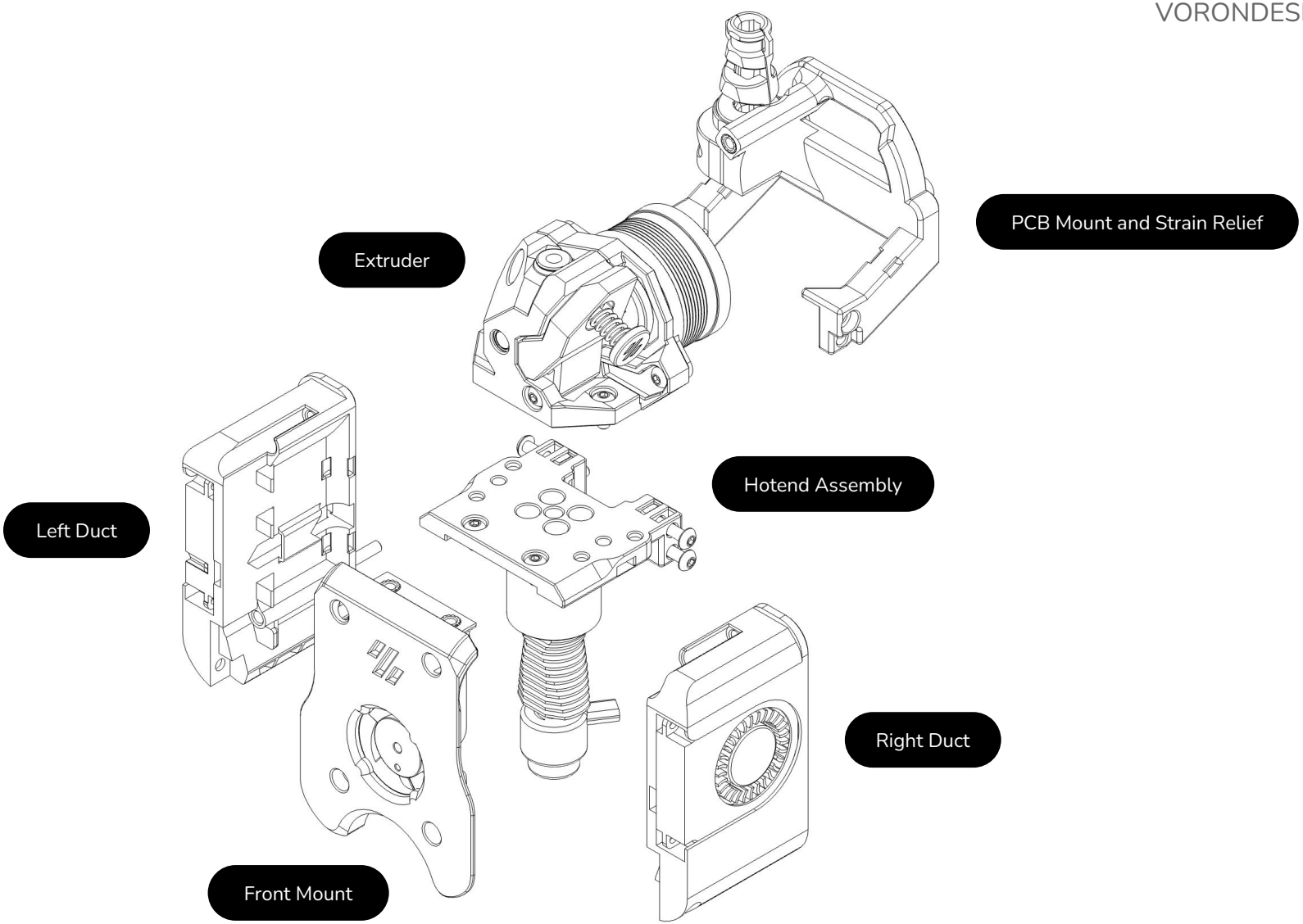


DOWNDRAFT

DownDraft is a modular, general-purpose toolhead system for Voron Trident, V2, Legacy, and Switchwire printers. It is designed to give builders more flexibility in component choice, cleaner cable routing, and easier long-term maintenance.

The system is available in two main variants: one for 4010 part cooling fans and one for 5015 part cooling fans. Both versions support 36 mm toolhead PCB boards, with support for StealthBurner-style boards where possible. The modular mounting system allows users to choose their preferred hotend, extruder, and cooling setup without changing the core toolhead structure. Adjustable hotend fan positions also make it easier to align airflow with different heatsink heights.

DownDraft includes an integrated LED light harness, built-in strain relief, and support for both umbilical and cable chain wiring. The design also keeps serviceability in mind, with better access to common maintenance areas so the toolhead can be inspected, cleared, and repaired with less disassembly.


PCB Mount and Strain Relief
Extruder
Hotend Assembly
Left Duct
Right Duct
Front Mount



4010 VERSION

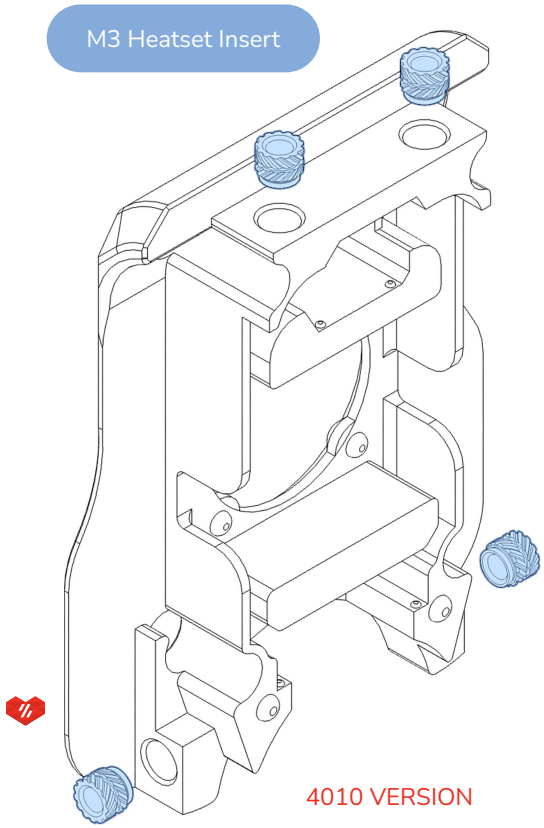


5015 VERSION

TOOLHEAD ASSEMBLY

The following pages include separate instructions for the 4010 and 5015 versions. Follow only the steps for the version you are building, then continue with the shared assembly steps when noted.

The front mount holds the 2510 hotend fan and LEDs, while the side fans provide part cooling in either the 4010 or 5015 configuration. The hotend block supports a range of common hotends and ties into the modular mounting system, allowing different extruder options to be used without changing the core toolhead structure.



4010 VERSION

FRONT

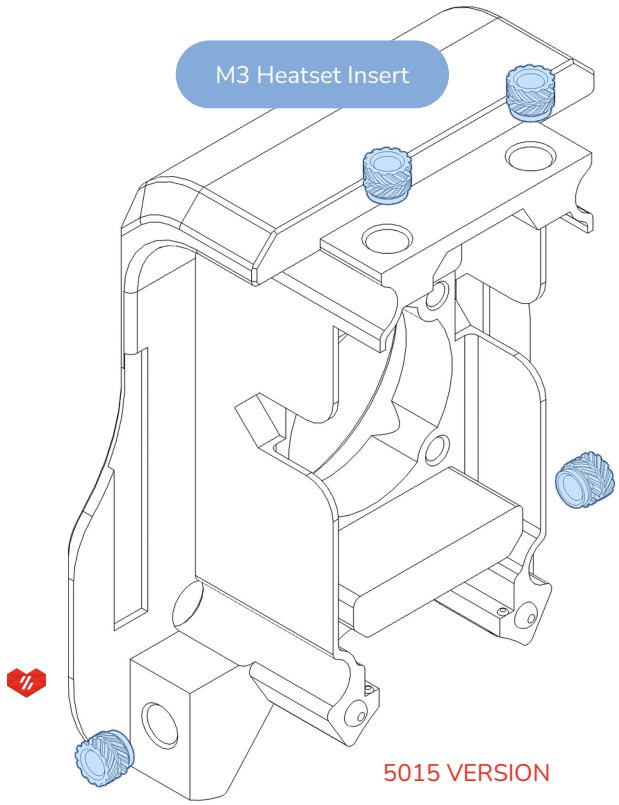
Install four heatset inserts into the front mount: two in the lower side holes and two in the top holes.

HEATSET INSERTS

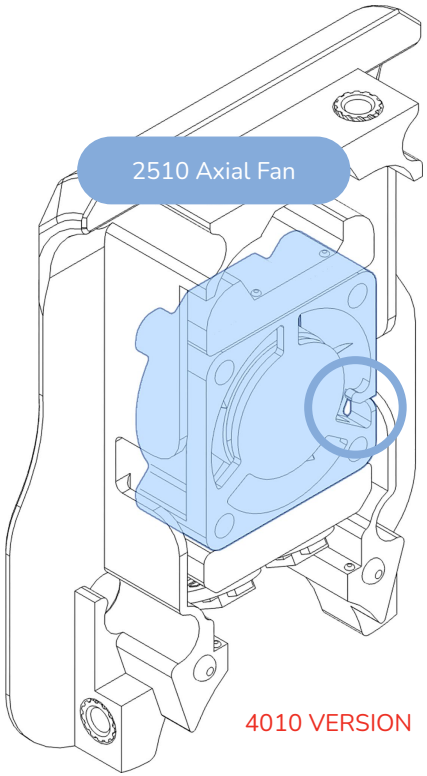
This design uses heatset inserts throughout the assembly. Make sure you are using the correct inserts before starting. Refer to the hardware reference for a close-up photo and the BOM for dimensions.

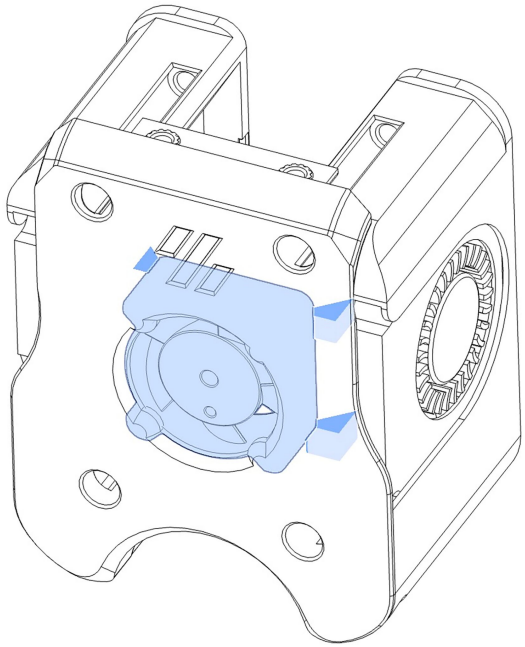
If you have not used heatset inserts before, we recommend reviewing the linked guide first.

https://voron.link/m5ybt4d

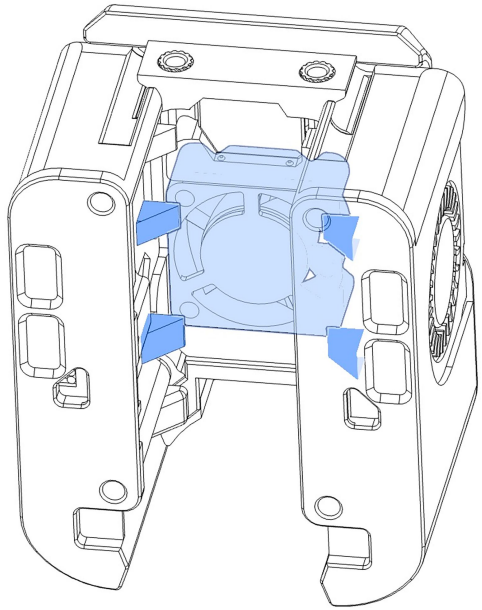


5015 VERSION





Note: Ducts are not installed yet. This visual shows the printed features that hold the 2510 axial fan in place.



HOTEND FAN

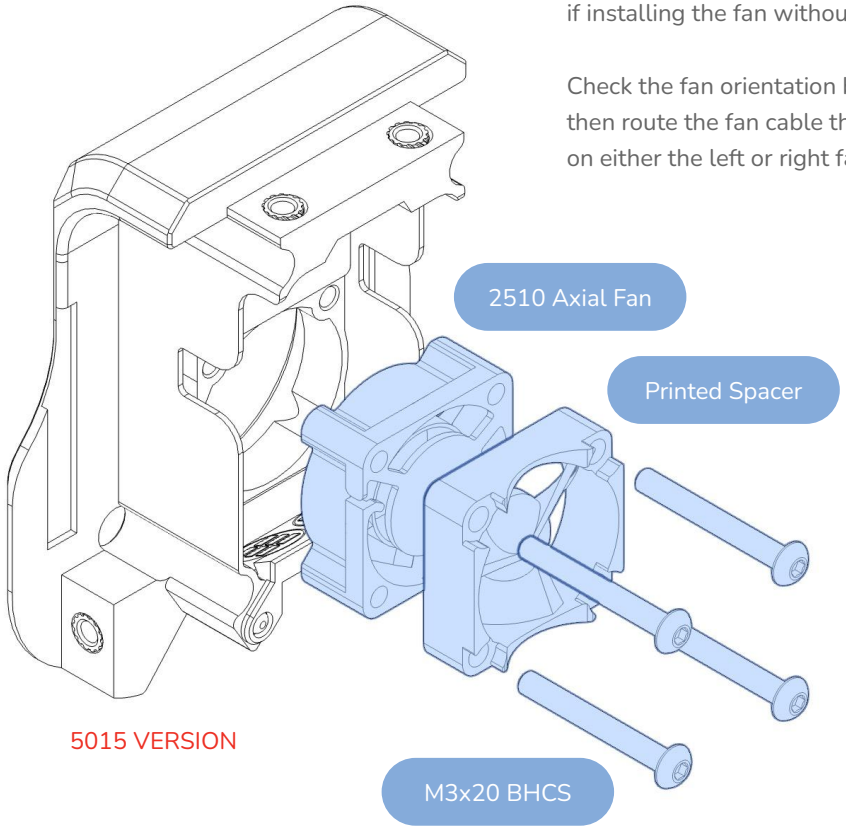
Insert the 2510 axial fan into the rear of the front mount. The fan will be held in place later by the left and right fan mounts.

Check the fan orientation before installing it. Position the fan cable so it can route through the cable duct on either the left or right fan mount.

HOTEND FAN + PRINTED SPACER

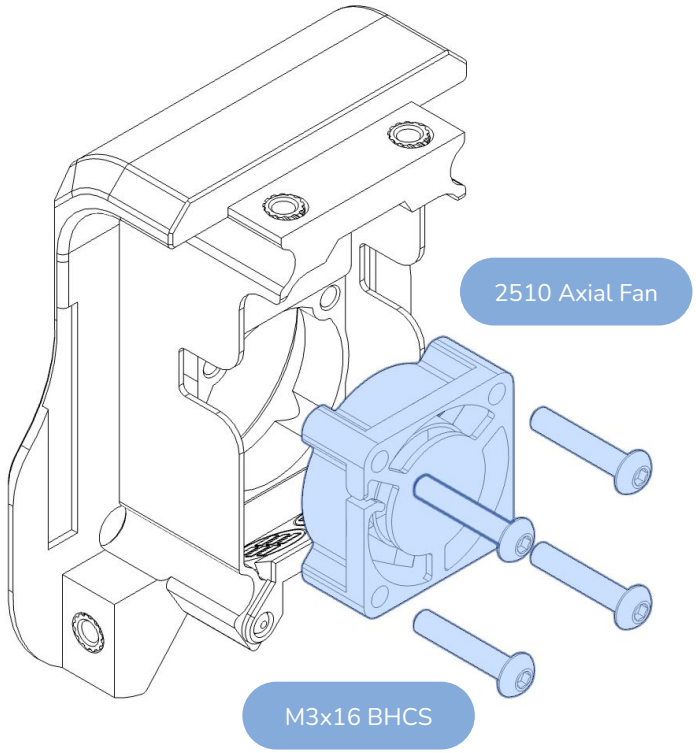
For the 5015 version, the hotend fan is screwed directly into the front mount. An optional printed spacer can be used to concentrate airflow toward the hotend. Use M3x20 BHCS if installing the printed spacer. Use M3x16 BHCS if installing the fan without the spacer.

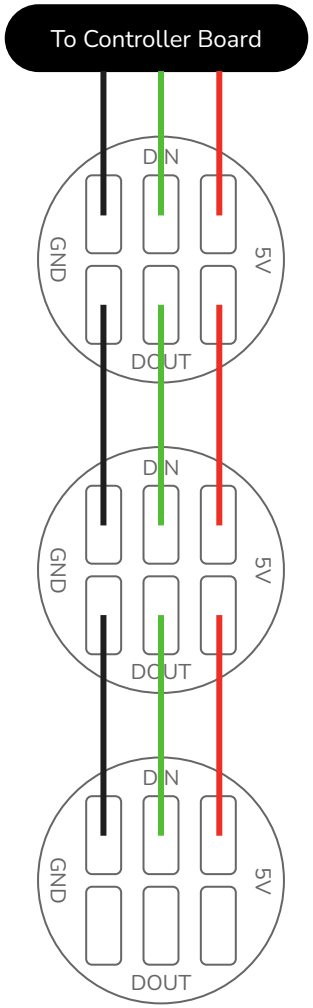
Check the fan orientation before installing it, then route the fan cable through the cable duct on either the left or right fan mount.



5015 VERSION

Note: Some 2510 axial fans, including the common Delta version, have smaller screw holes. For these fans, you may install M2.5 heatset inserts and secure the fan to the front part with M2.5 screws.

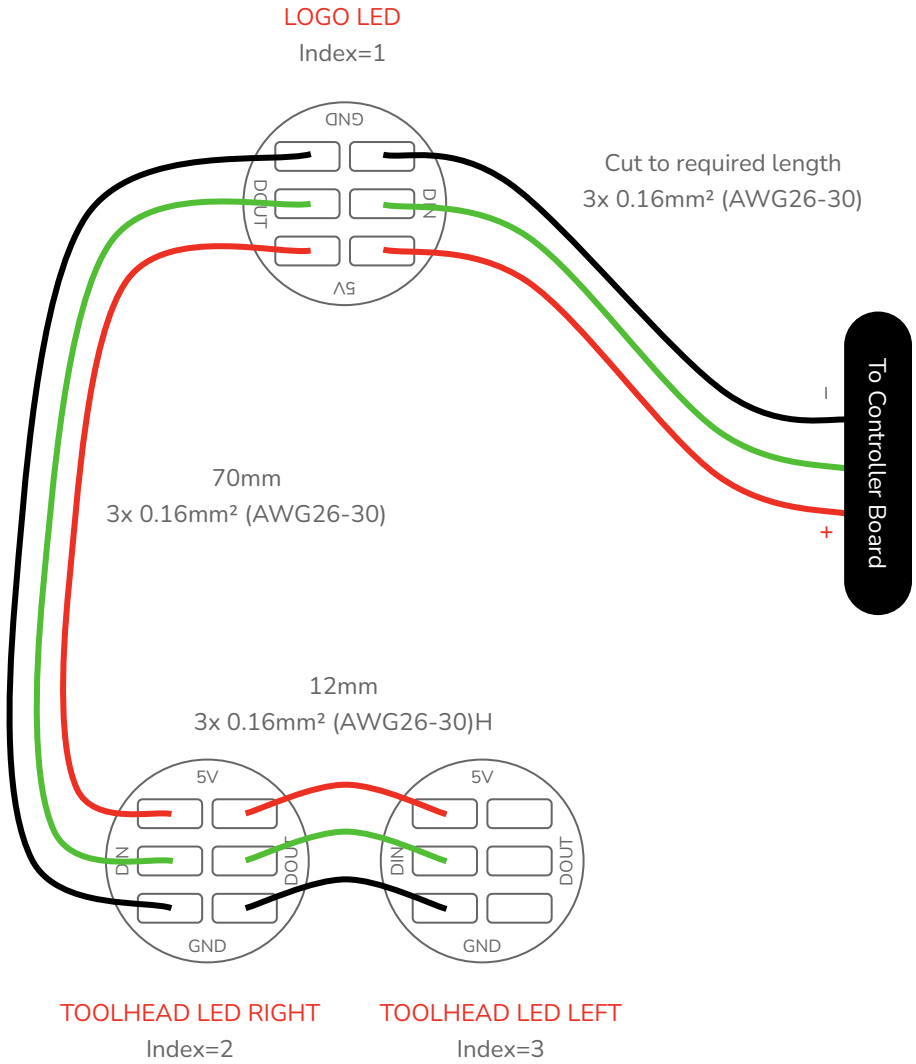




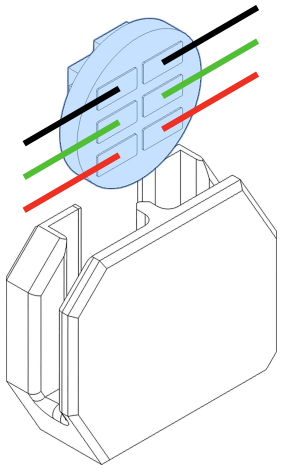
LED WIRING

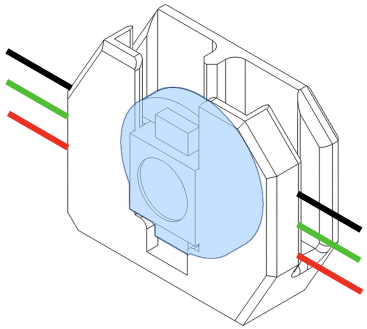
Wire the LEDs in series, connecting the DOUT pad from one LED to the DIN pad on the next LED.

Follow the wiring path shown on the right. The logo LED is first in the chain, followed by the right toolhead LED, then the left toolhead LED.



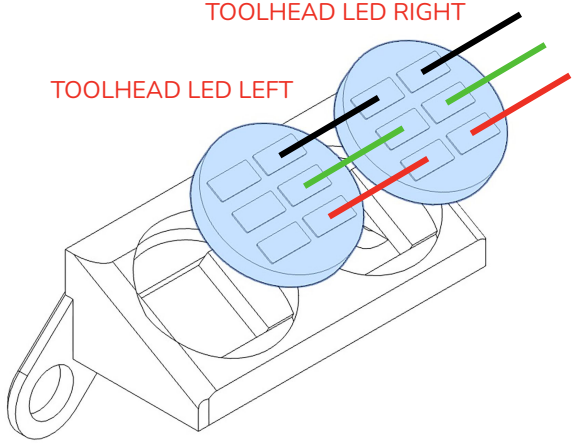
LOGO LED

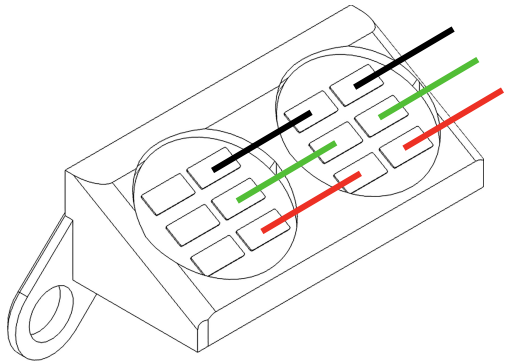


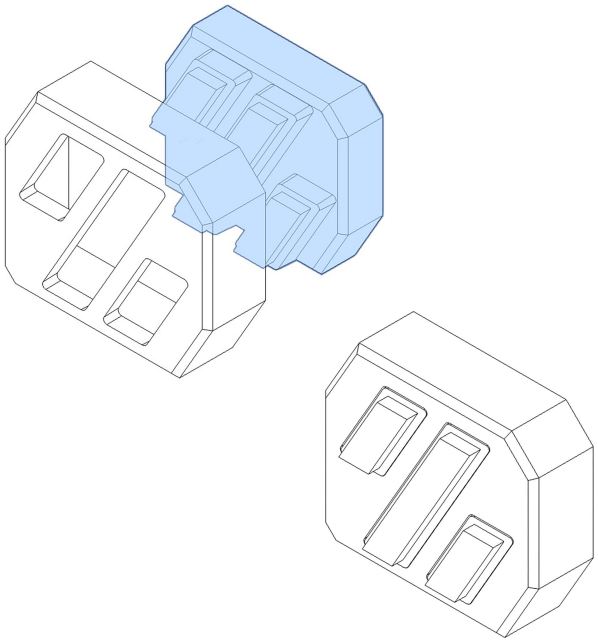


TOOLHEAD LED RIGHT

TOOLHEAD LED LEFT







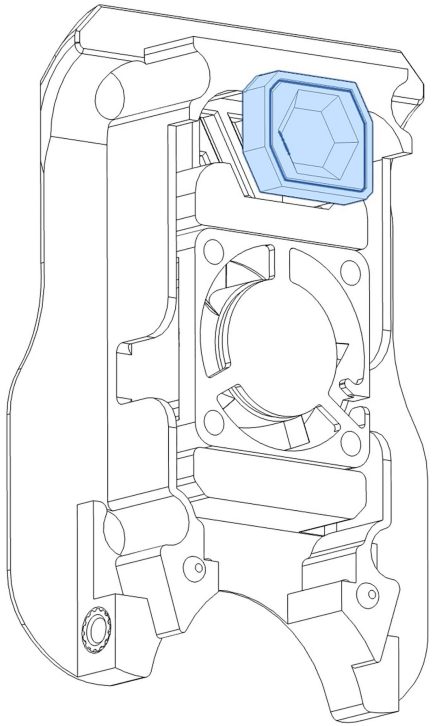
LED DIFFUSER AND MASK

Print the diffuser (highlighted) in translucent filament to help spread the light evenly.

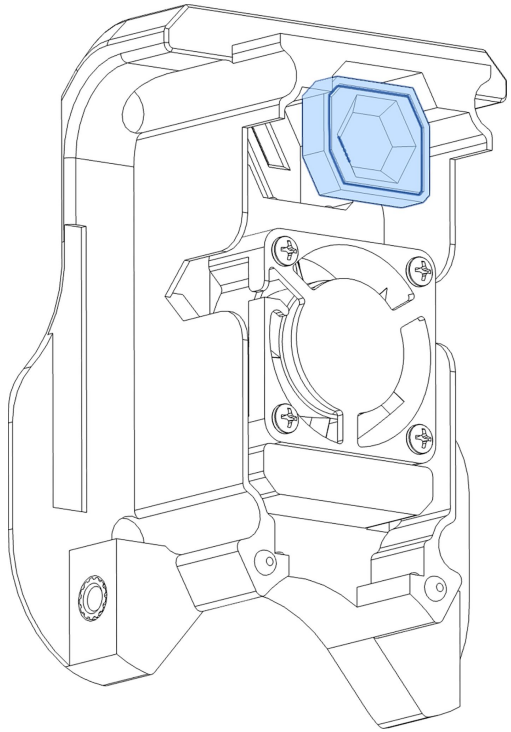
Print the mask in opaque filament to block light bleed and keep the logo edges crisp.

DIFFUSER INSERTION

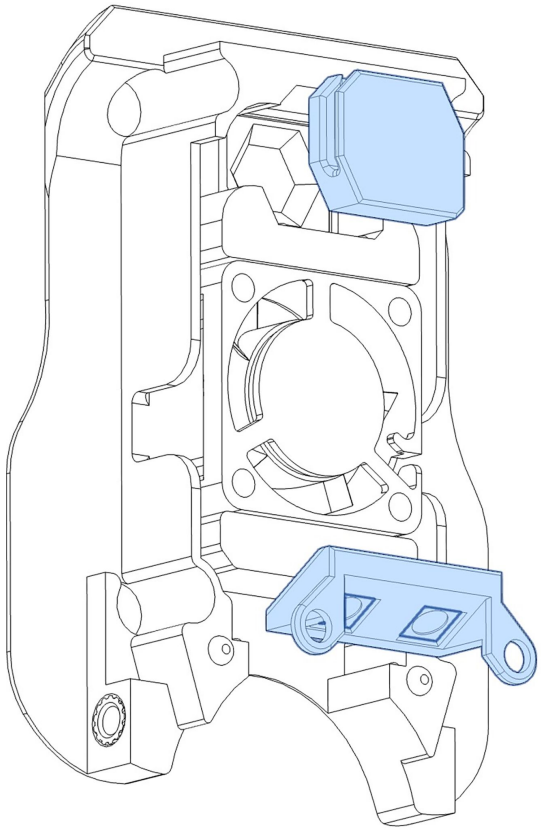
Insert the diffuser and mask, then slide both parts forward until they are fully seated.



4010 VERSION



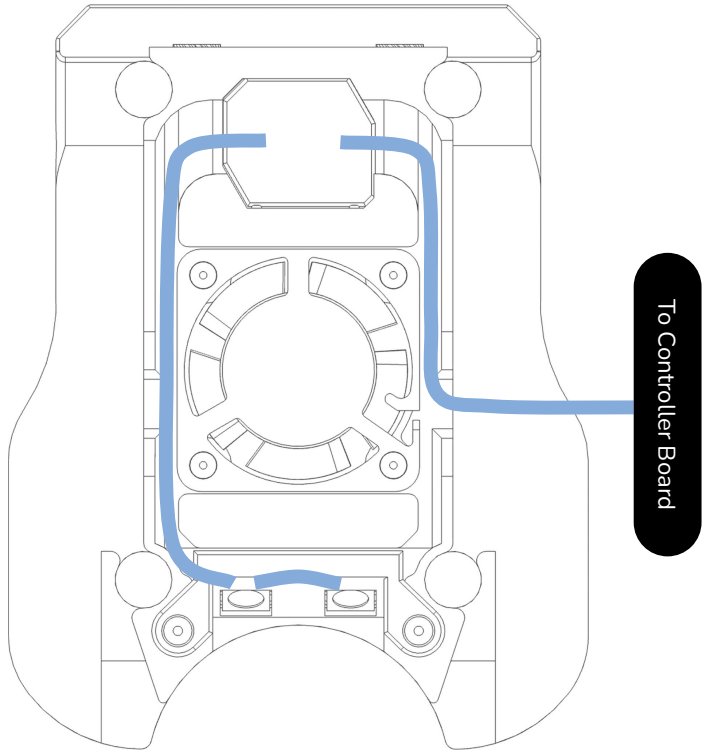
5015 VERSION

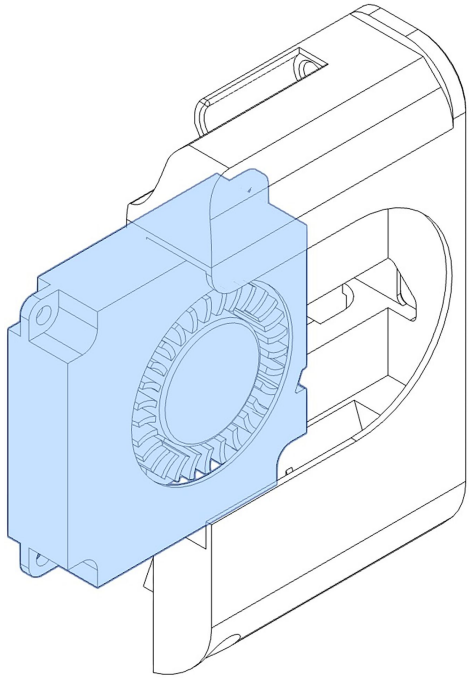


SIMILAR FOR
BOTH VERSIONS

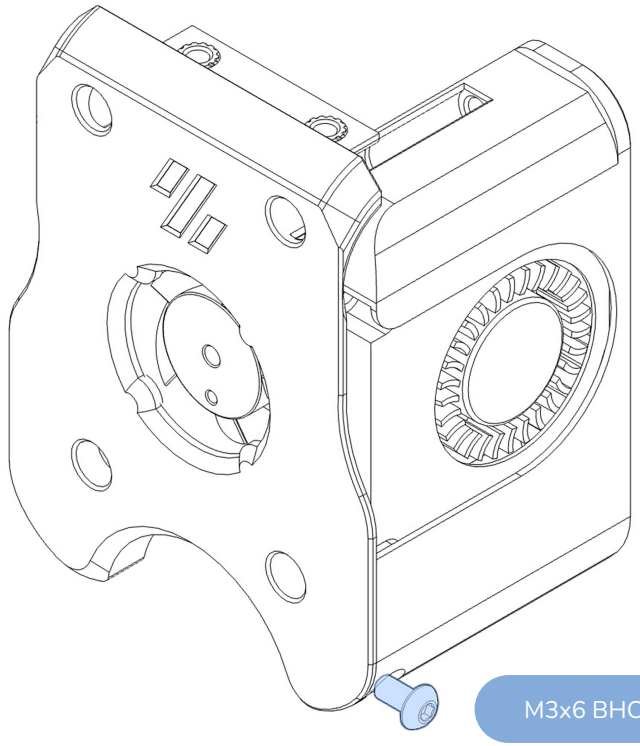
LEDS INSERTION

Install the wired LEDs and carriers, then route the wires through the wire channels.





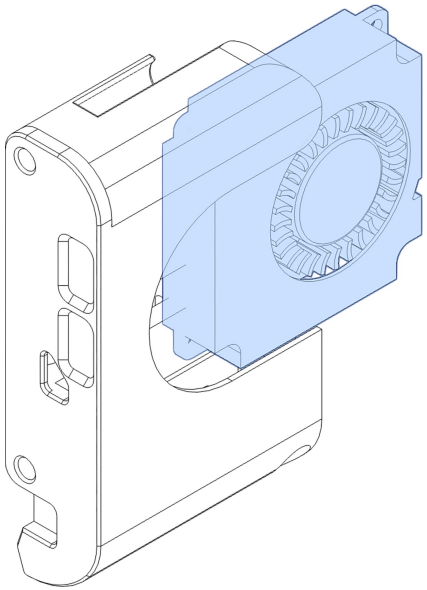
4010 VERSION

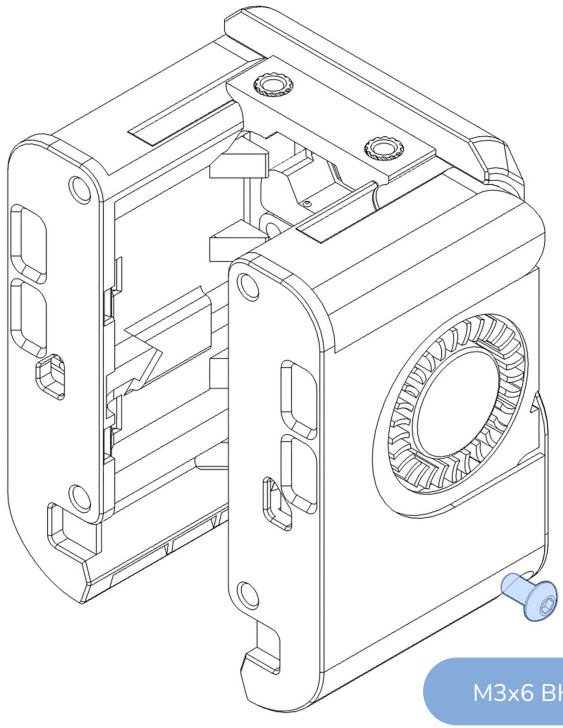


4010 FAN DUCTS

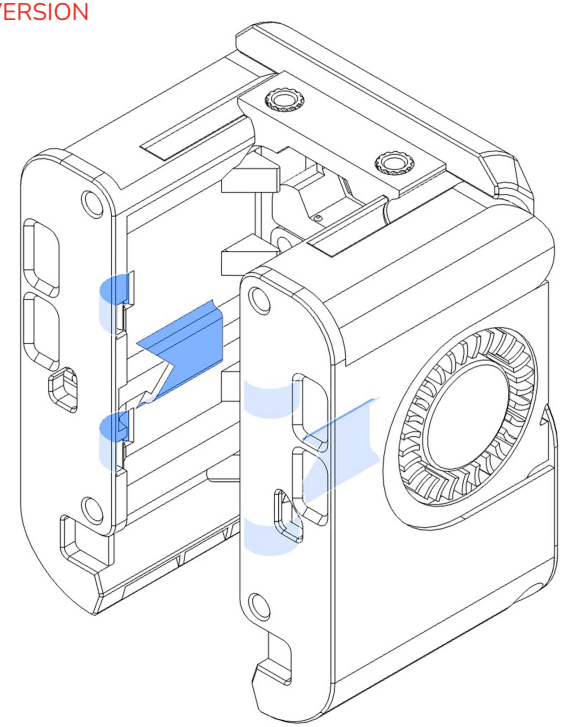
Install the 4010 fan into the fan duct, then route the fan cable through the cable channel.

Secure the duct assembly to the front assembly with one M3x6 BHCS.





4010 VERSION



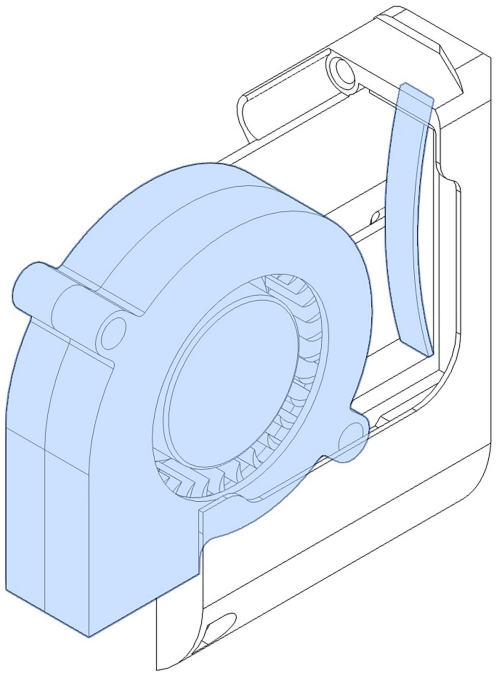
4010 FAN LEFT DUCT

Insert the 4010 fan into the left duct, then route the fan cable through the cable channel.

For the left fan, you may need to remove the cable from the retainer on the fan body. Take care not to pull the wires from the fan PCB.

Secure the duct assembly to the front assembly with one M3x6 BHCS.

Optional: insert zip ties through the holes on the inside rear of the fan mounts for additional cable management.

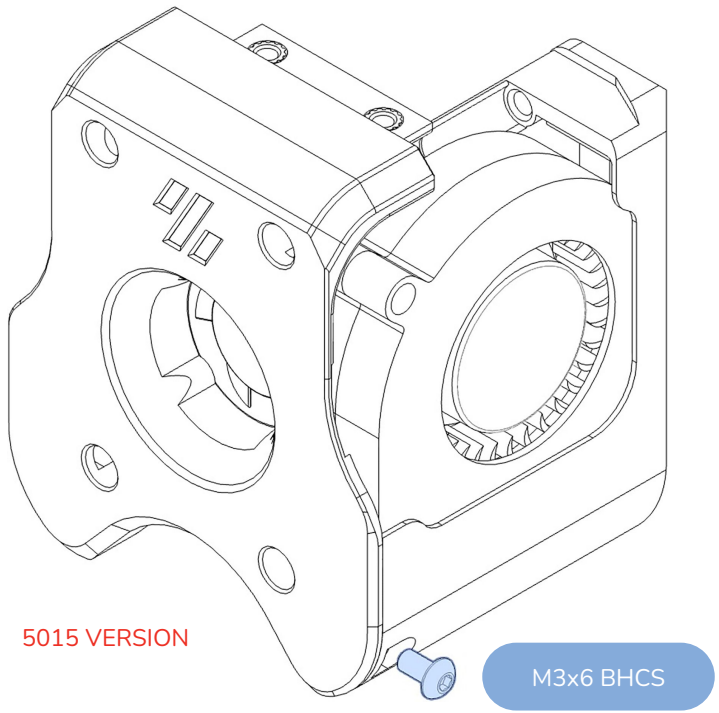


5015 FAN RIGHT DUCT

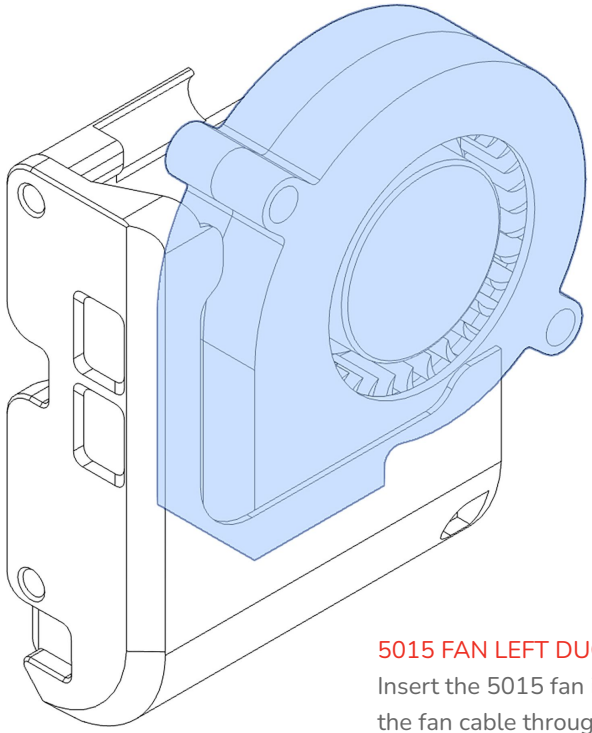
Install 5015_Fan_Spring_x2 inside the right duct, then insert the 5015 fan. Route the fan cable through the cable channel.

Secure the duct assembly to the front assembly with one M3x6 BHCS.

Note: Do not install the 5015_Fan_Spring parts when using slightly oversized 5215 fans, such as Berserker Vindrs. There is not enough clearance for the springs with these fans.



5015 VERSION

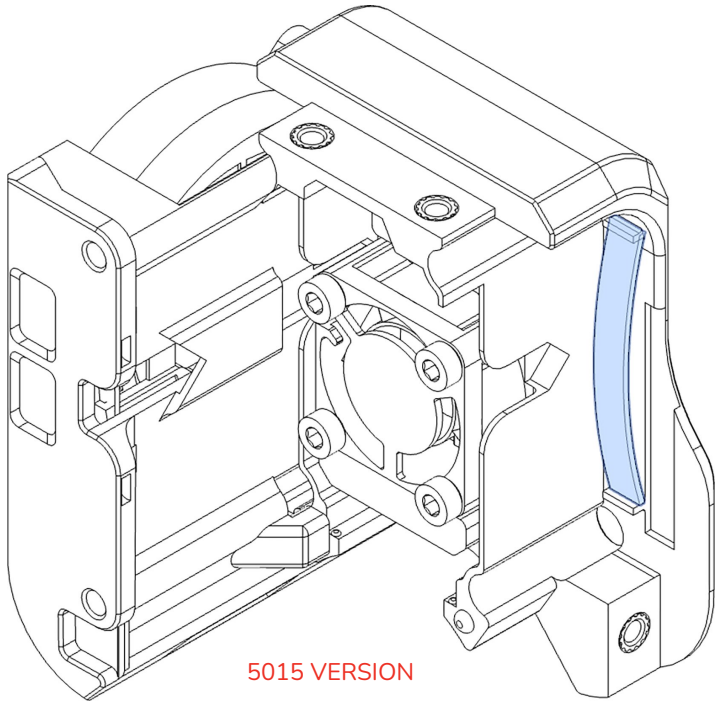


5015 FAN LEFT DUCT

Insert the 5015 fan into the left duct, then route the fan cable through the cable channel.

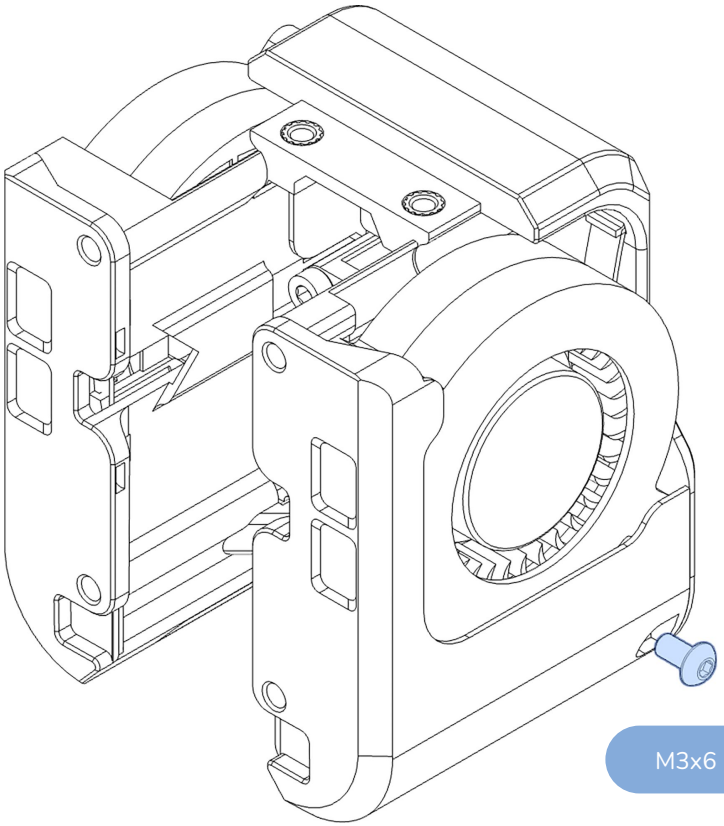
For the left fan, you may need to remove the cable from the retainer on the fan body. Take care not to pull the wires from the fan PCB.

Viewed from the rear, install 5015_Fan_Spring_x2 inside the front assembly.



5015 VERSION

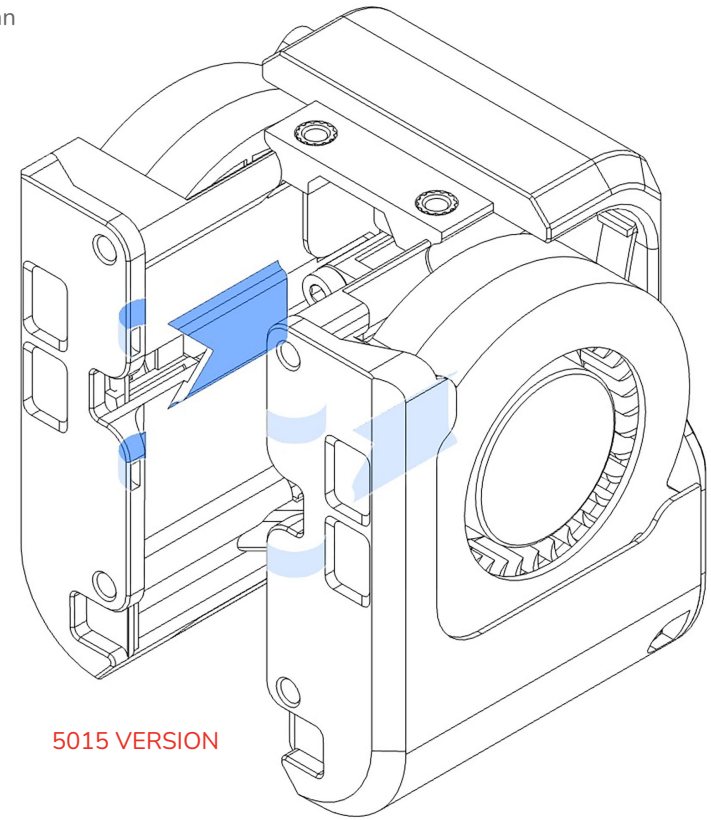
Note: Do not install the 5015_Fan_Spring parts when using slightly oversized 5215 fans, such as Berserker Vindrs. There is not enough clearance for the springs with these fans.



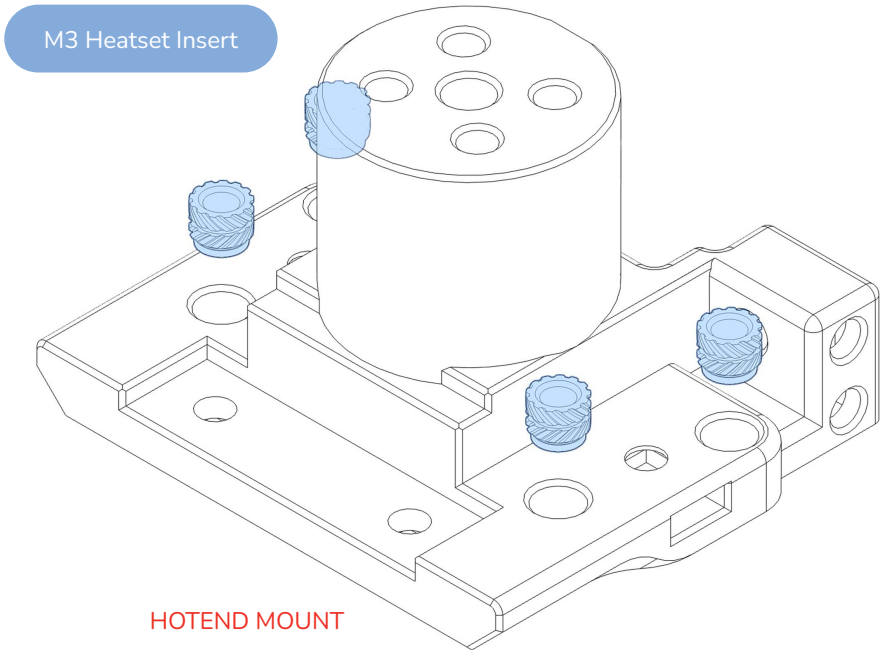
5015 FAN LEFT DUCT

Secure the duct assembly to the front assembly with one M3x6 BHCS.

Optional: insert zip ties through the holes on the inside rear of the fan mounts for additional cable management.



5015 VERSION

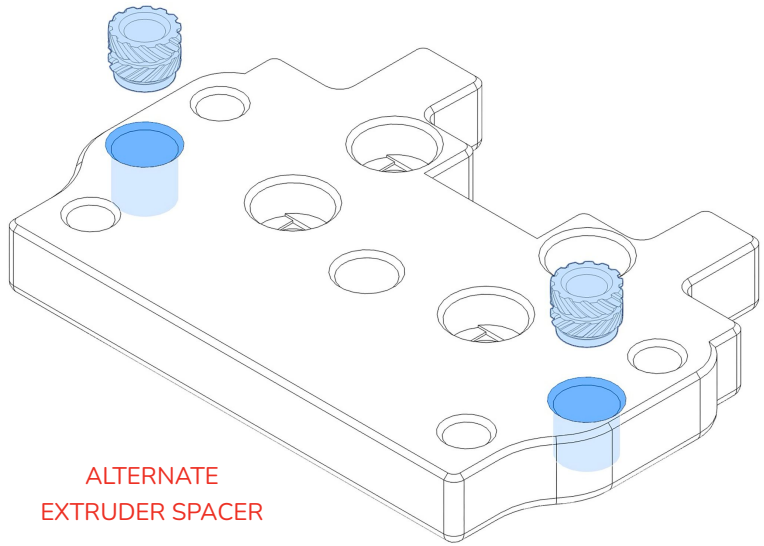


HOTEND MOUNT

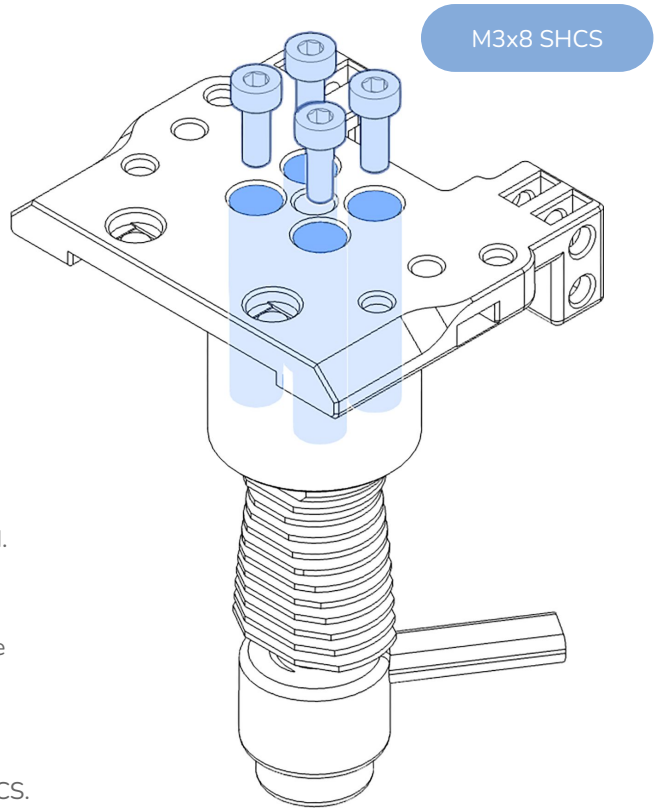
NON-ORBITER V2 MOUNTING

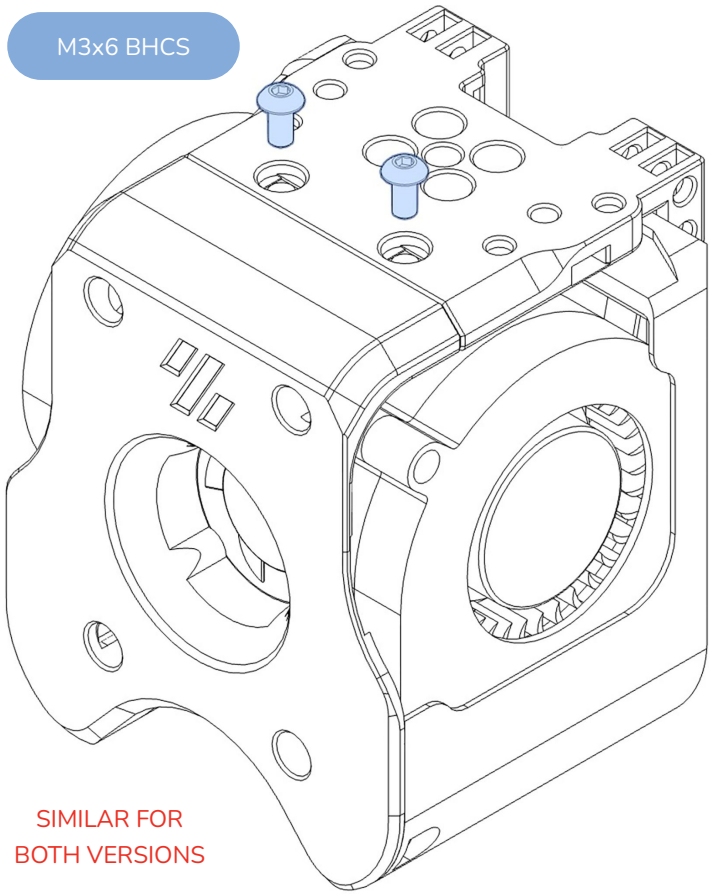
For extruders that do not use the Orbiter v2 mounting pattern, install four heatset inserts into the hotend/extruder mount from below.

Then, install two heatset inserts into the Alternative_Extruder_Spacer.



ALTERNATE
EXTRUDER SPACER





SIMILAR FOR
BOTH VERSIONS

HOTEND

Attach the hotend to the hotend mount using four M3x8 SHCS or M2.5x8 SHCS, depending on your hotend. Position the hotend so the cables face the rear of the toolhead.

Optional: add zip ties to the rear of the hotend mount to help retain the hotend wiring.

Secure the hotend assembly to the front assembly with two M3x6 BHCS.

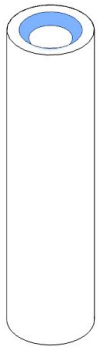
Route the cables behind the hotend, then up and out the rear of the toolhead. Make sure the cables will not be pinched between the toolhead and carriage mount.

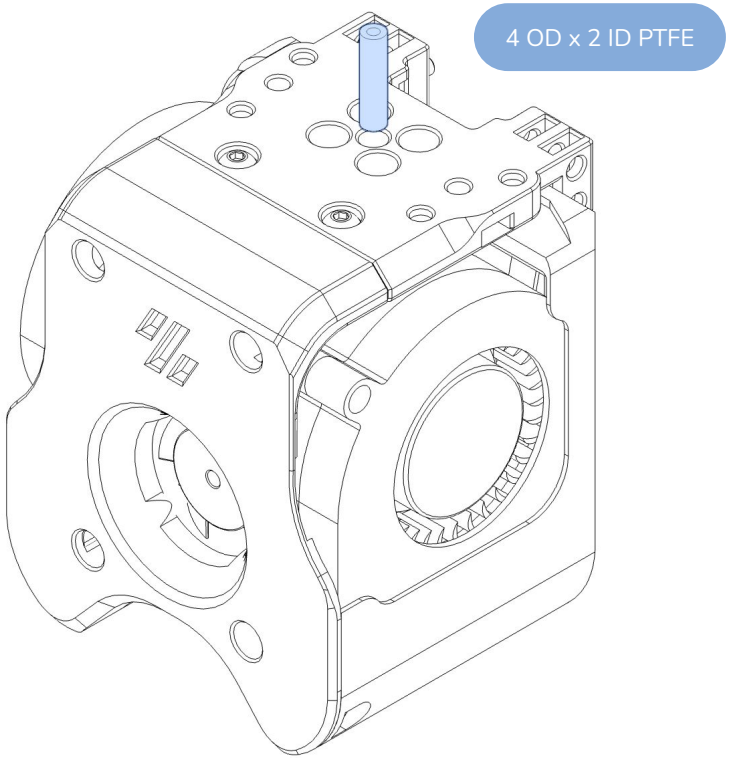
PTFE STICKOUT

Install the 4 mm OD x 2 mm ID PTFE tube so it extends above the toolhead far enough to seal against the extruder.

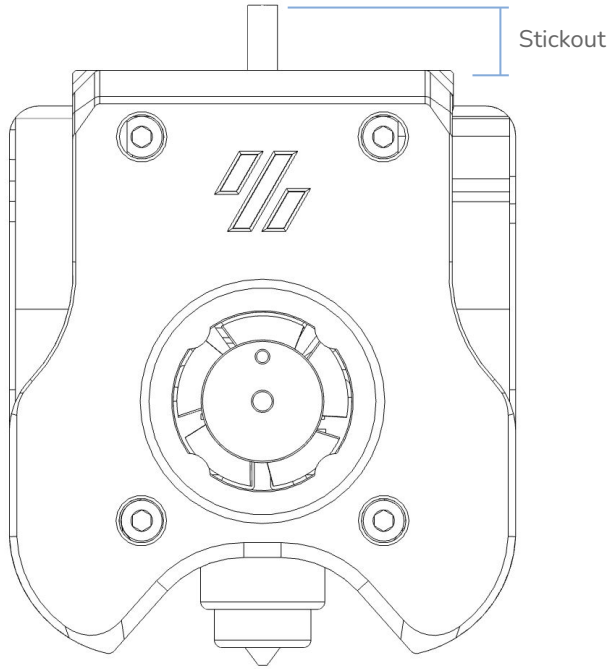
The required stickout will vary depending on your hotend and extruder combination.

Optional: Lightly chamfer the inside edge of the PTFE tube entry with a drill bit to make filament insertion easier.





SIMILAR FOR
BOTH VERSIONS



ORBITER V2 MOUNTING

Extruders that use the Orbiter v2 mounting pattern attach directly to the hotend mount.

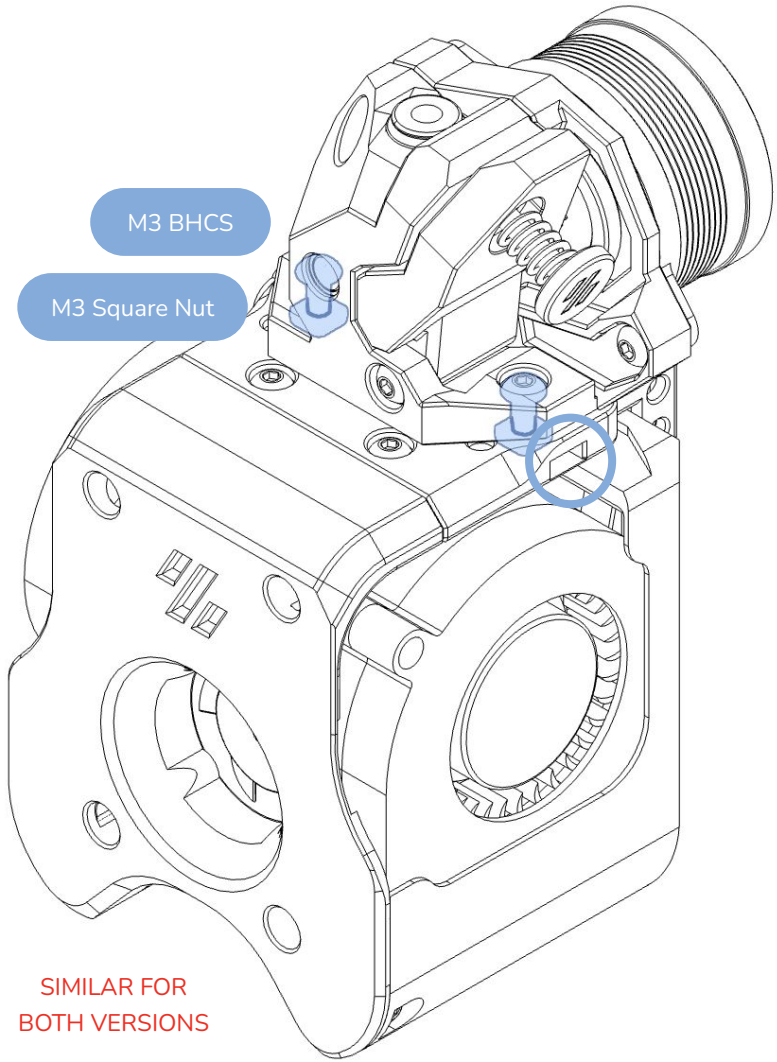
Insert one M3 square nut into the slot (circled), then secure that side of the extruder with an M3 screw. Repeat for the other side.

EXTRUDER MOUNTING SCREWS

The required screw length depends on the extruder being used.

Choose M3 screws long enough to fully engage the M3 square nuts, but not so long that they protrude through the bottom of the hotend mount.

Note: M3 square nuts are used to secure the extruder because they can be tightened with more torque than heatset inserts. This helps reduce toolhead “nod” caused by extruder forces.



SIMILAR FOR BOTH VERSIONS

NON-ORBITER V2 MOUNTING

For extruders that do not use the Orbiter v2 mounting pattern, first attach the Alternative_Extruder_Spacer to the toolhead using four M3x8 BHCS.

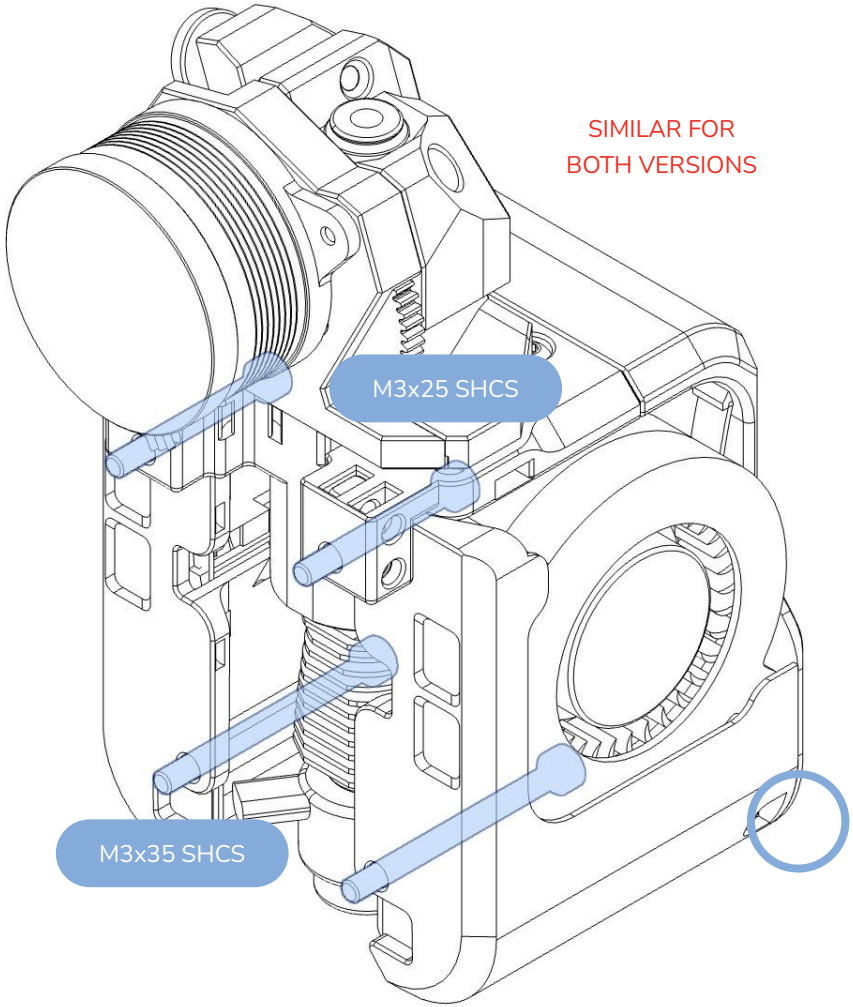
The extruder can then be mounted to the Alternative_Extruder_Spacer.

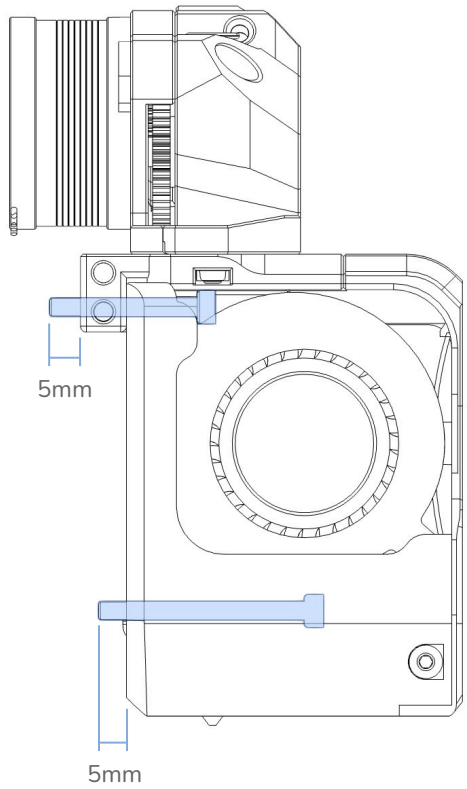
Note: For extruders that use the LGX Lite mounting pattern, attach the extruder to the Alternative_Extruder_Spacer before installing that assembly onto the toolhead.

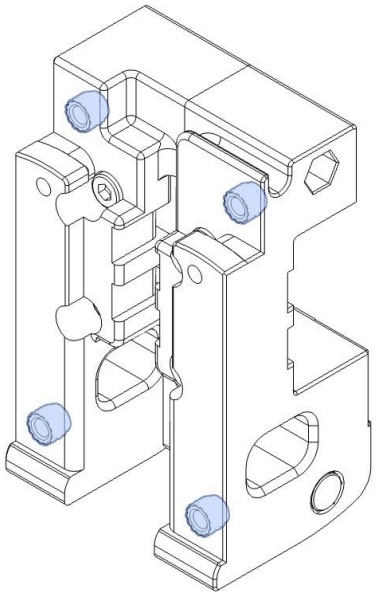
TOOLHEAD MOUNTING SCREWS

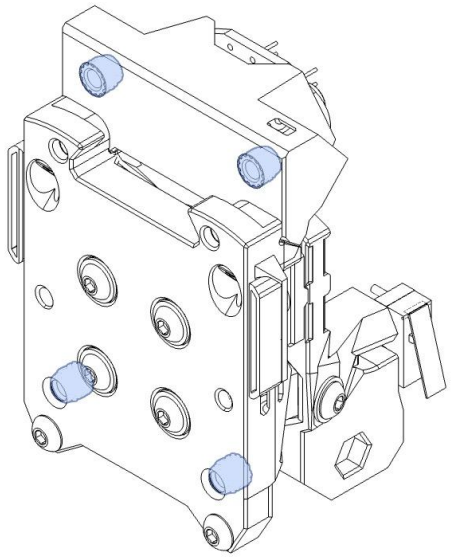
Insert the two lower M3x35 SHCS from the front. The screws should protrude about 5 mm from the rear of the toolhead.

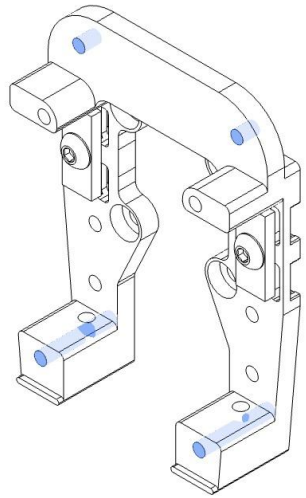
Insert the two upper M3x25 SHCS from the front. These screws need to pass by the 5015 fans. If there is not enough clearance to slide the upper screws past the fans, loosen the lower side screws on the toolhead (circled). Push the mounting screws past the fans, then re-tighten the side screws.











MOUNTING COMPATIBILITY

Downdraft uses the CW2 toolhead mounting pattern. This allows the assembled toolhead to mount directly to a standard CW2 carriage, as well as Tap and other CNC carriage mounts designed for StealthBurner compatibility.

TOOLHEAD MOUNTING

The toolhead is secured to the carriage using the four mounting screws inserted from the front (see previous page).

Make sure the toolhead is seated flat against the carriage before tightening the screws in an X pattern.

TOOLHEAD MOUNTS

Several mounts are available for attaching hardware to the rear of the toolhead. These parts can be found in the toolhead parts directories.

Available mount options include:

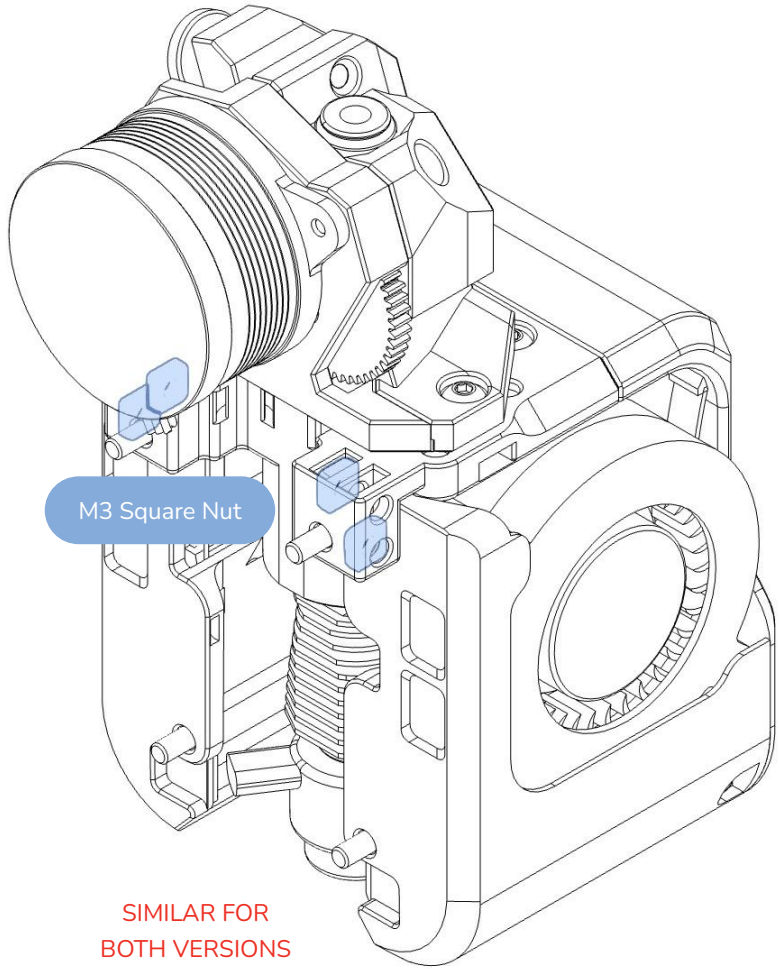
- Cable chain mounts
- Various toolhead PCB mounts
- Generic ADXL board mount
- PUG umbilical mount

NOTE: Some of these provided mount options are extruder specific and may not fit if a spacer is placed between the toolhead and the extruder, e.g. a filament cutter

We have included two examples:

- A **cable chain option** if your toolhead wiring will be routed through a cable chain, either without a toolhead PCB or with a standard non-CANBUS breakout PCB.
- A **PCB + PUG umbilical option** if your build uses a toolhead PCB with a CANBUS umbilical and PUG strain relief.

The examples show the build steps for two common setups. Your exact steps may vary based on your component choices.



SIMILAR FOR
BOTH VERSIONS

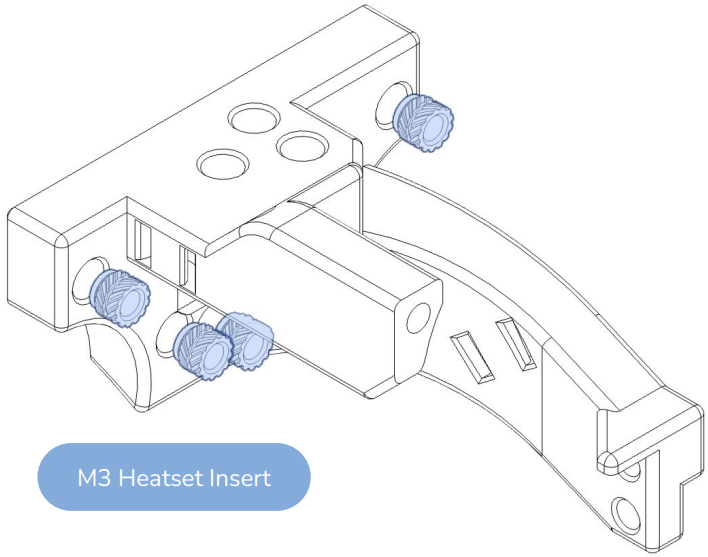
M3 SQUARE NUTS

To attach hardware, M3 square nuts are used in the rear slots of the hotend mount as shown.

Some mounts use only one side, while others use both sides. Keep the toolhead level during installs so the M3 square nuts do not slip out.

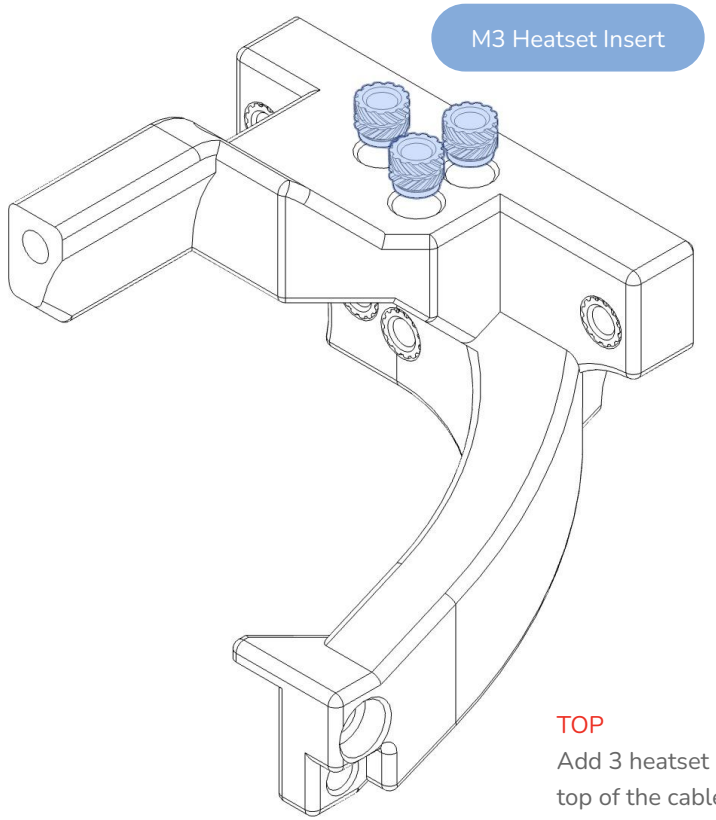
CABLE CHAIN OPTION

This section covers the cable chain wiring option for builds not using a PCB, or using a standard non-CANBUS breakout toolhead PCB. In this setup, the toolhead wiring is routed through a cable chain secured to the toolhead with a printed mount.



FRONT

Add 4 heatset inserts to the front of the cable chain mount.



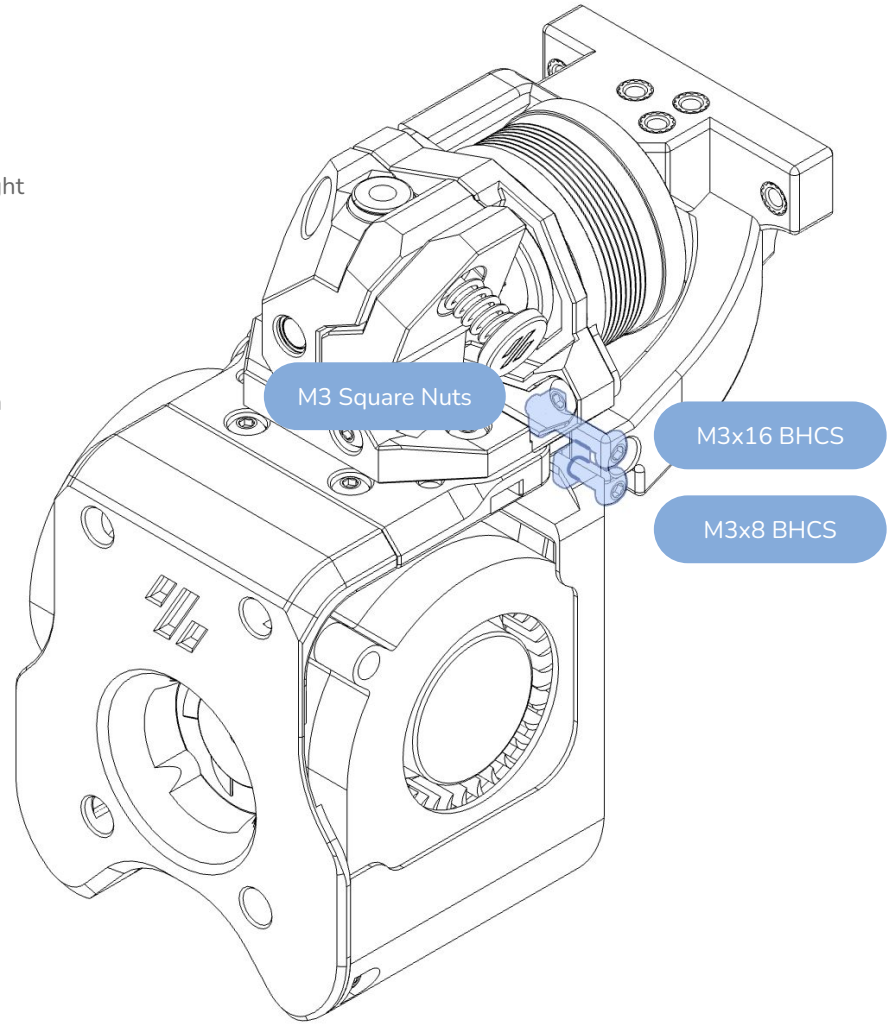
TOP

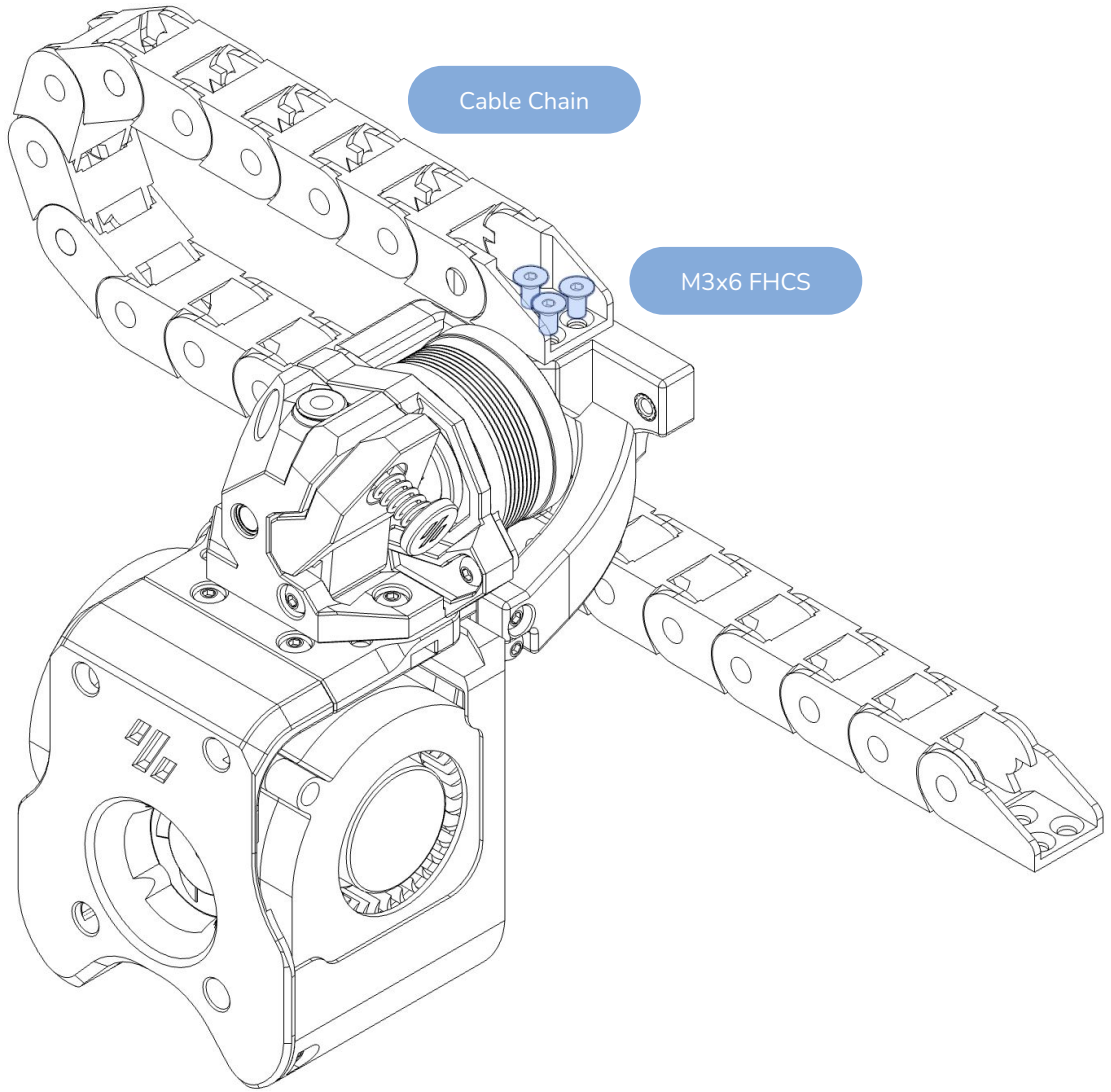
Add 3 heatset inserts to the top of the cable chain mount.

CABLE CHAIN MOUNT

The cable chain mount uses only the right side of M3 square nuts.

Secure the mount to the toolhead using an M3x8mm BHCS screw in the lower hole on the right side at the rear of the toolhead into the M3 Square nuts. Then use an M3x16mm BHCS screw in the upper hole.

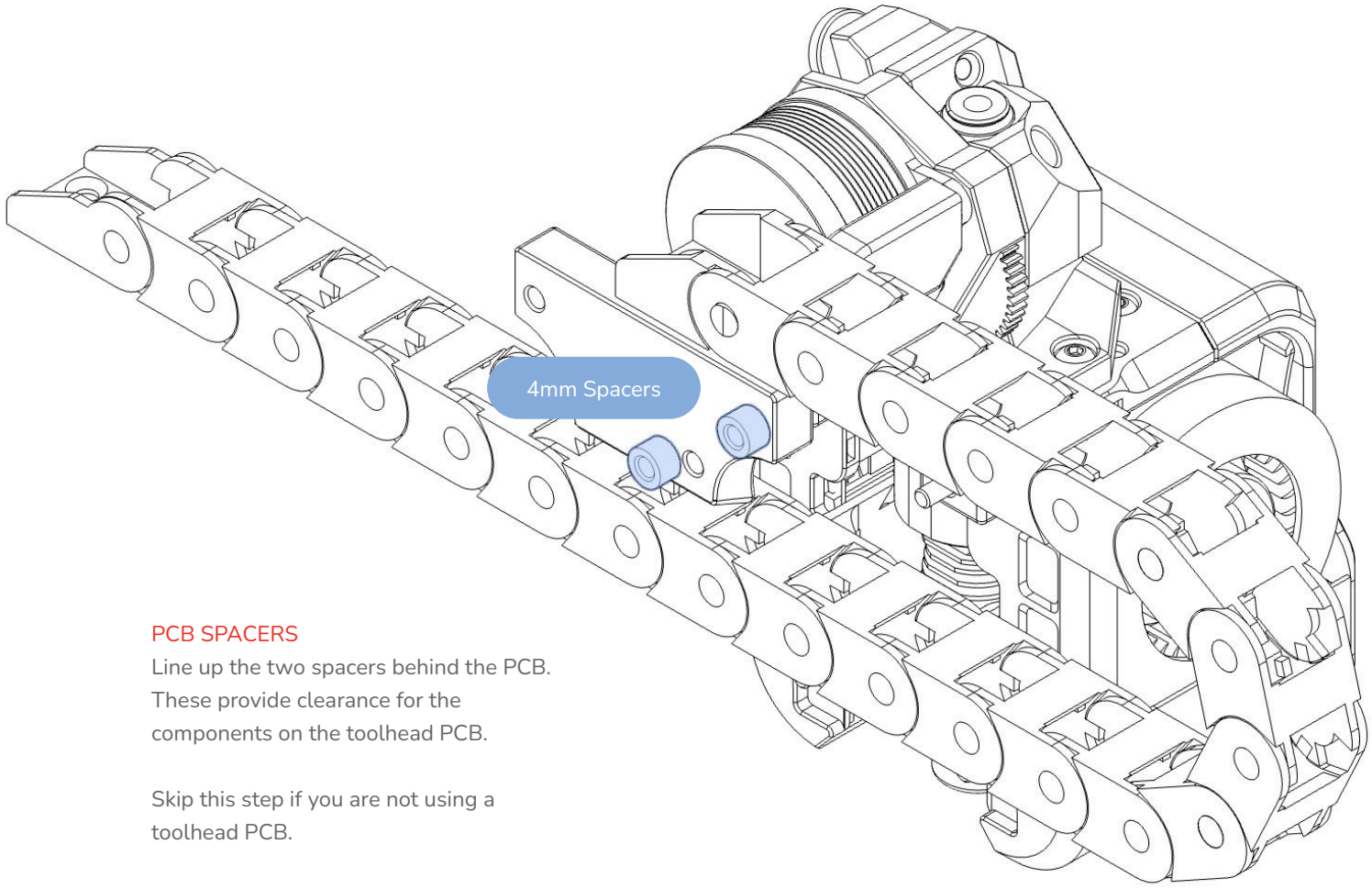




CABLE CHAIN

Secure the cable chain with M3x6 FHCS. Some cable chains use two screws, others use three screws.

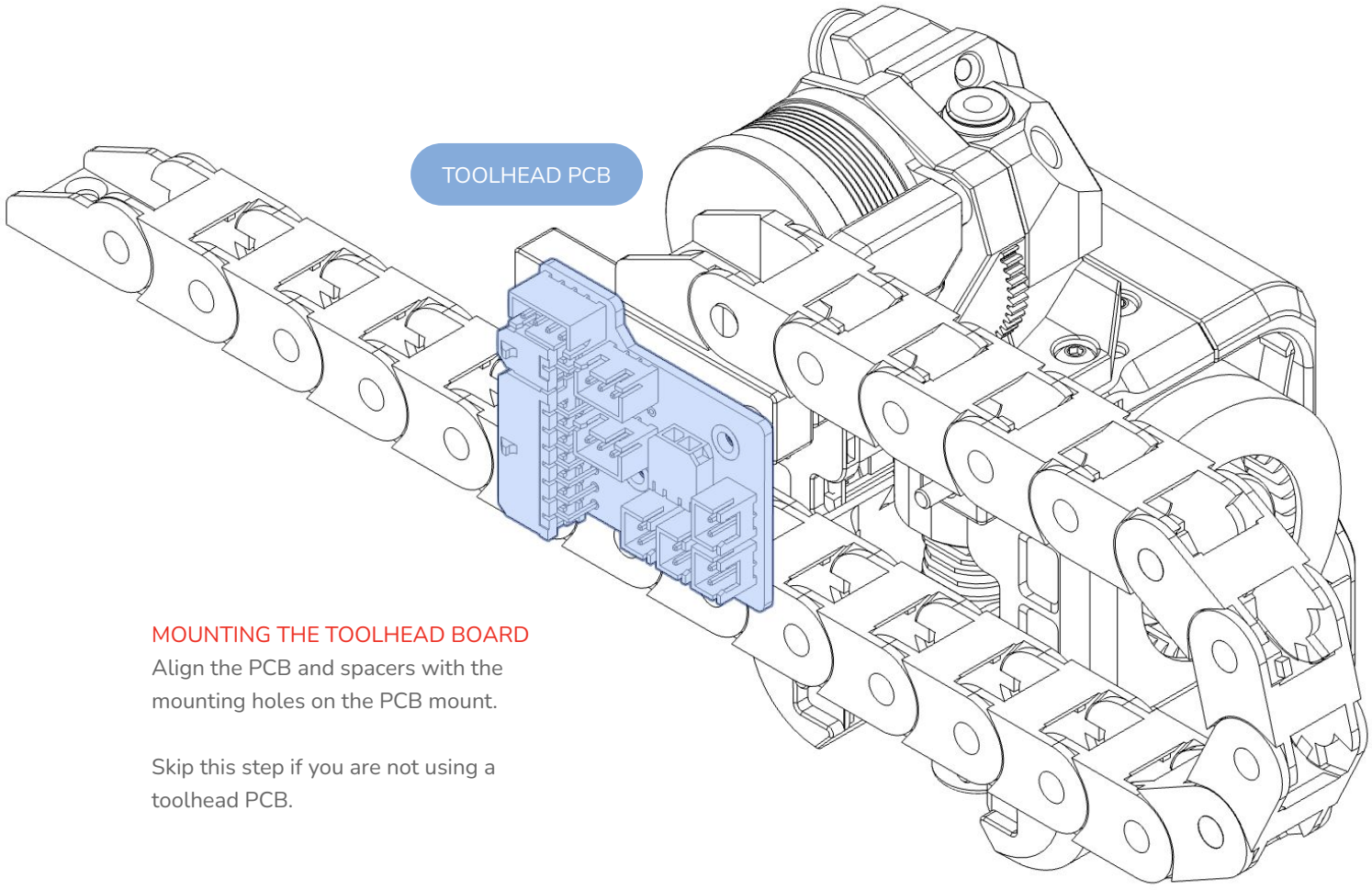
For a proper cable chain bend, use the fixed end link at the toolhead side, and use the unfixed end link opposite the toolhead.



PCB SPACERS

Line up the two spacers behind the PCB. These provide clearance for the components on the toolhead PCB.

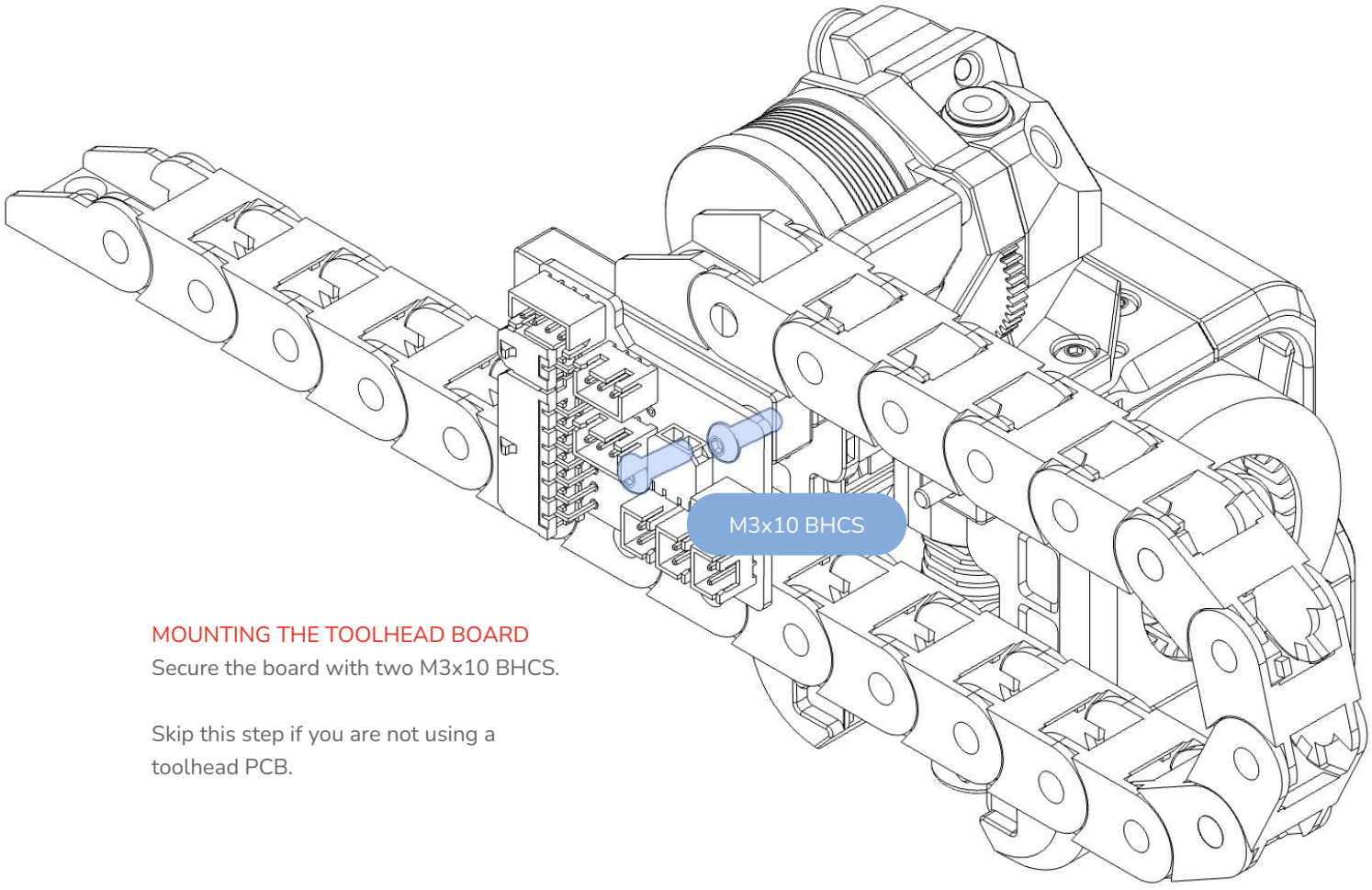
Skip this step if you are not using a toolhead PCB.



MOUNTING THE TOOLHEAD BOARD

Align the PCB and spacers with the mounting holes on the PCB mount.

Skip this step if you are not using a toolhead PCB.



MOUNTING THE TOOLHEAD BOARD

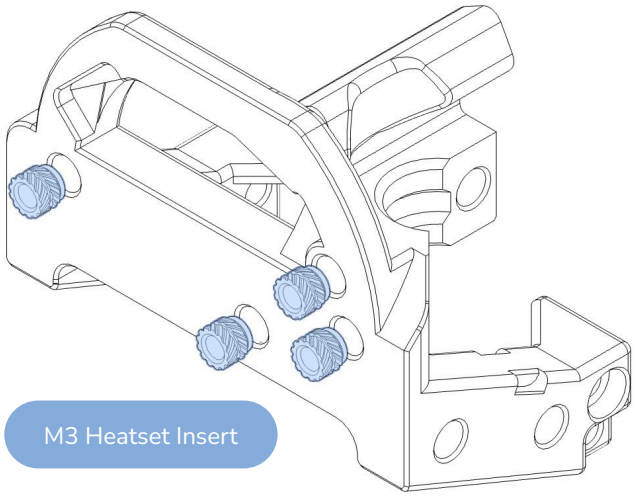
Secure the board with two M3x10 BHCS.

Skip this step if you are not using a toolhead PCB.

PCB + PUG UMBILICAL OPTION

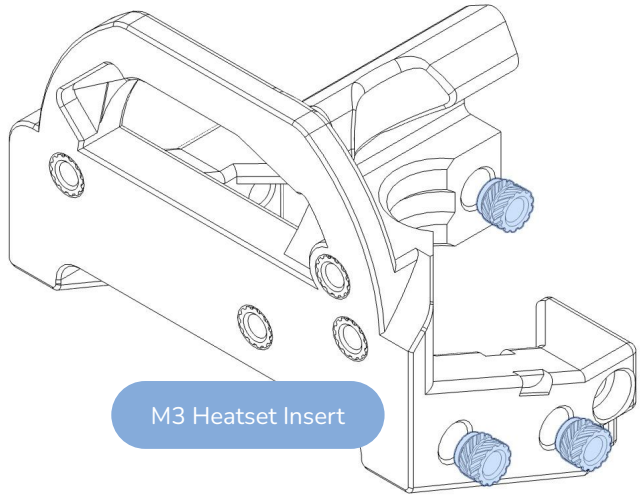
This section covers the umbilical wiring option using a toolhead PCB and PUG strain relief system*. The PUG captures and supports the CANBUS wire before it enters the toolhead, helping reduce movement at the connector and keeping the cable path controlled. This setup is intended for cleaner umbilical routing while still keeping the PCB, wiring, and common service areas accessible.

* PUG (Parametric umbilical gland) by fbeauKmi (www.printables.com/model/378567-pug-parametric-umbilical-gland)



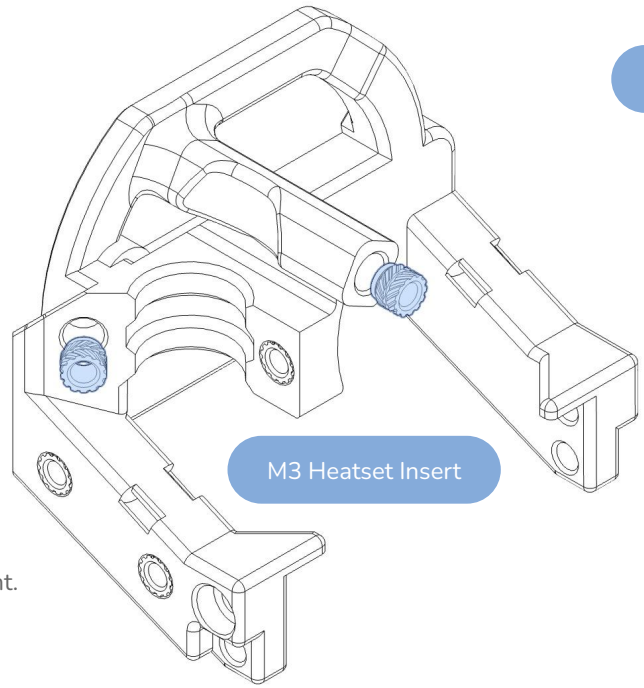
BACK

Add 4 heatset inserts to the back of the PUG/chain mount.



LEFT

Add 2 heatset inserts to the left of the PUG/chain mount.



FRONT

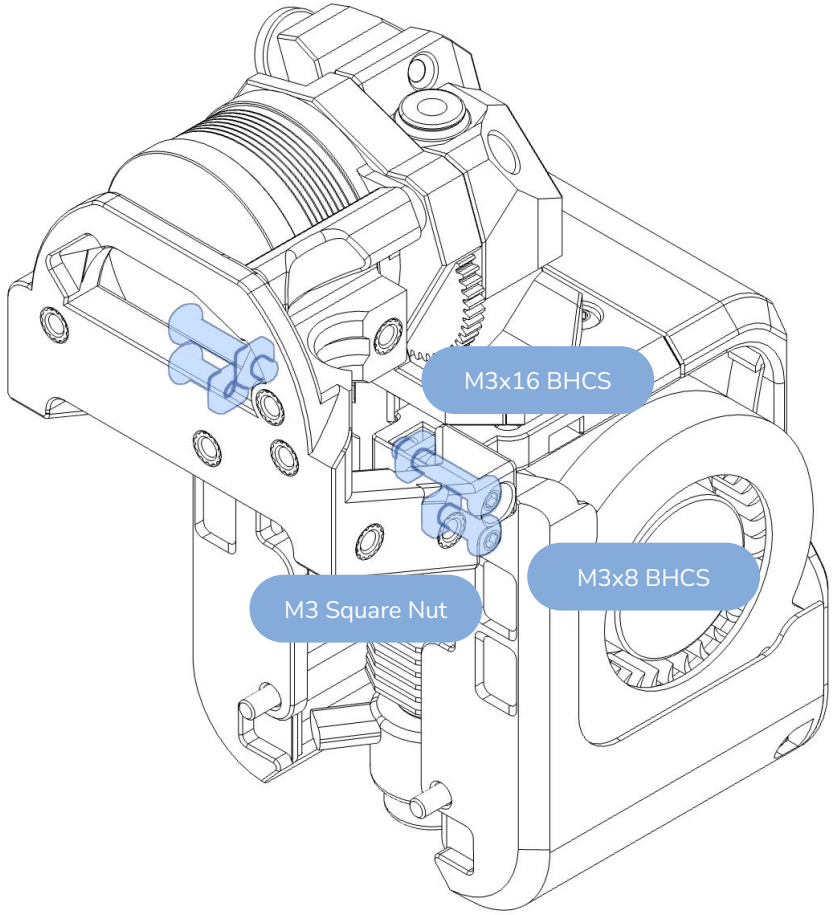
Add 2 heatset inserts to the front of the PUG/chain mount.

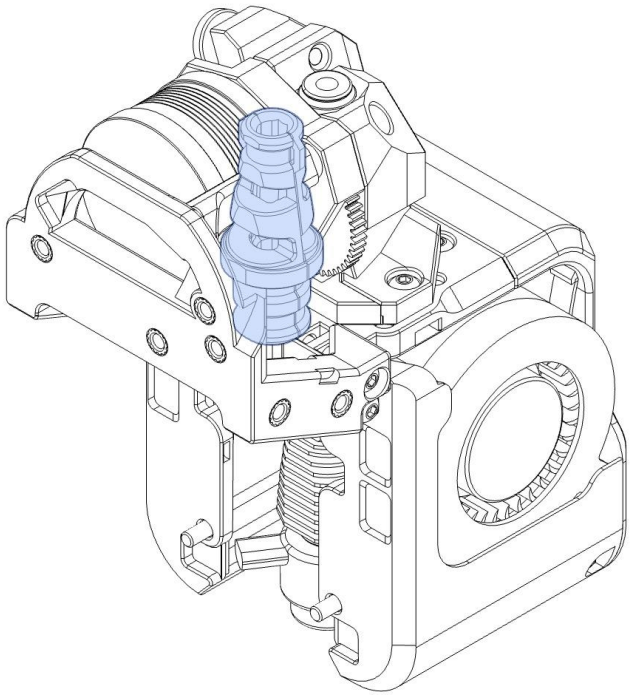
PCB + PUG UMBILICAL MOUNT

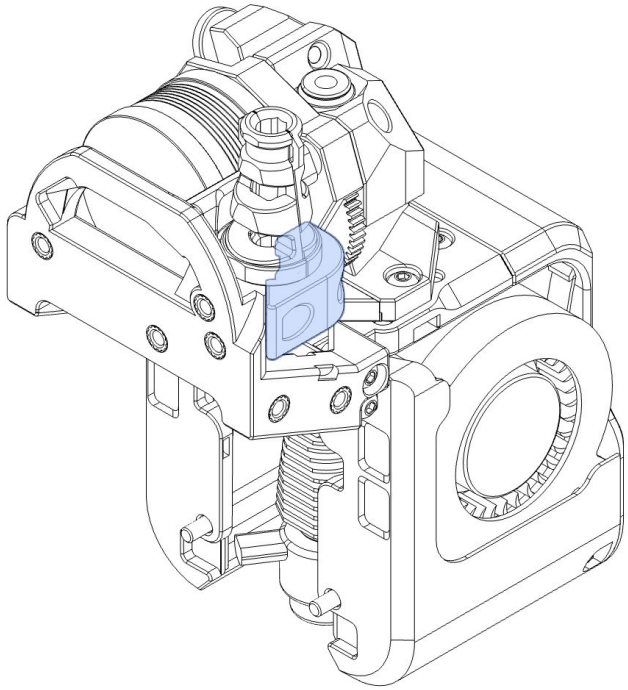
The PCB + PUG umbilical mount uses M3 square nuts on both sides of the hotend mount.

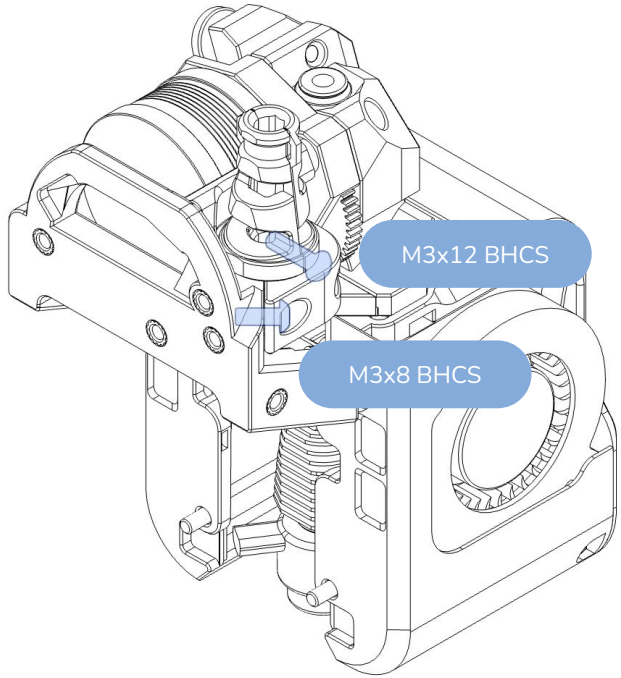
Secure the toolhead PCB mount to the rear of the toolhead using two M3x8 BHCS in the lower left and right mounting holes. These fasten into the M3 square nuts mentioned earlier.

Then secure the upper mounting holes with two M3x16 BHCS to M3 square nuts.







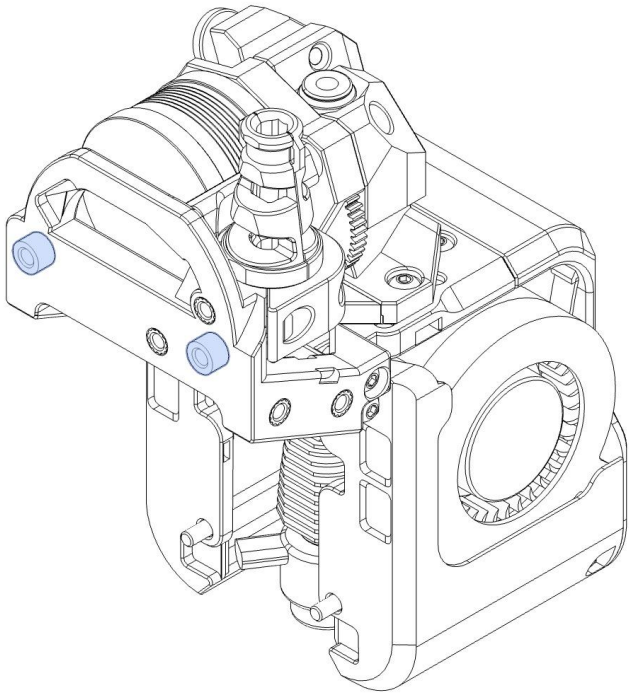


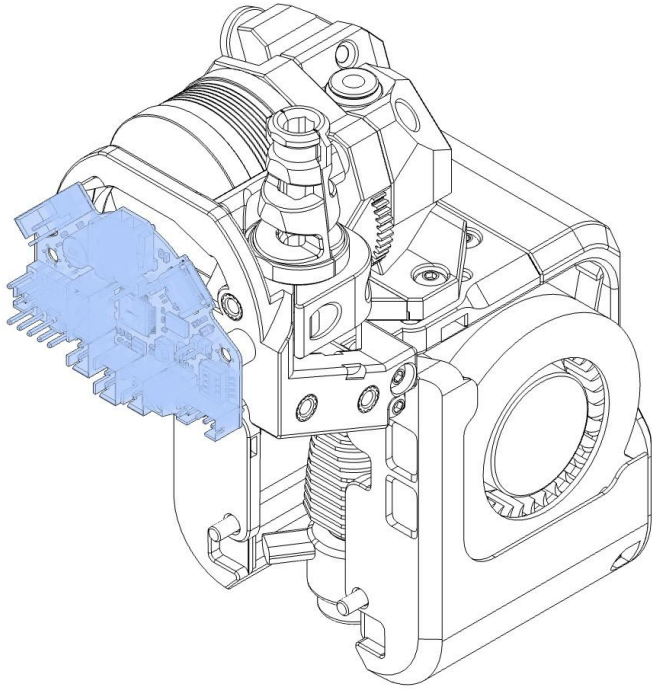
PUG MOUNT

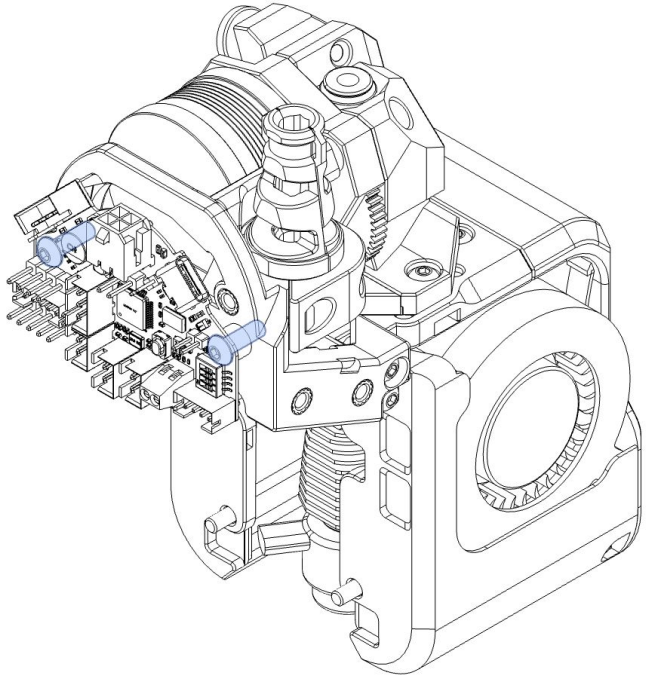
Place the CANBUS wire between the two halves of the PUG, then seat the assembled PUG into the mount. The grooves on the PUG should rest inside the mount.

PUG CLAMP

Install the clamp over the PUG and secure it with one M3x8 BHCS and one M3x12 BHCS.







PCB SPACERS

Line up the two spacers behind the PCB. These provide clearance for the components on the toolhead PCB.

MOUNTING THE TOOLHEAD BOARD

Align the PCB and spacers with the mounting holes on the PCB mount, then secure the board with two M3x10 BHCS.

HOW TO GET HELP

If you need assistance with your build, we're here to help. Head on over to our Discord group and post your questions. This is our primary medium to help VORON Users and we have a great community that can help you out if you get stuck. Alternatively, you can use our subreddit.



https://discord.gg/voron



https://www.reddit.com/r/VORONDesign

REPORTING ISSUES

Should you find an issue in this document or have a suggestion for an improvement please consider opening an issue on GitHub (https://github.com/VoronDesign/Downdraft/issues).

When raising an issue please include the relevant page numbers and a short description; annotated screenshots are also very welcome.

We periodically update the manual based on the feedback we get.

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VORON



We build space shuttles with gardening tools
so anyone can have a space shuttle of their own.